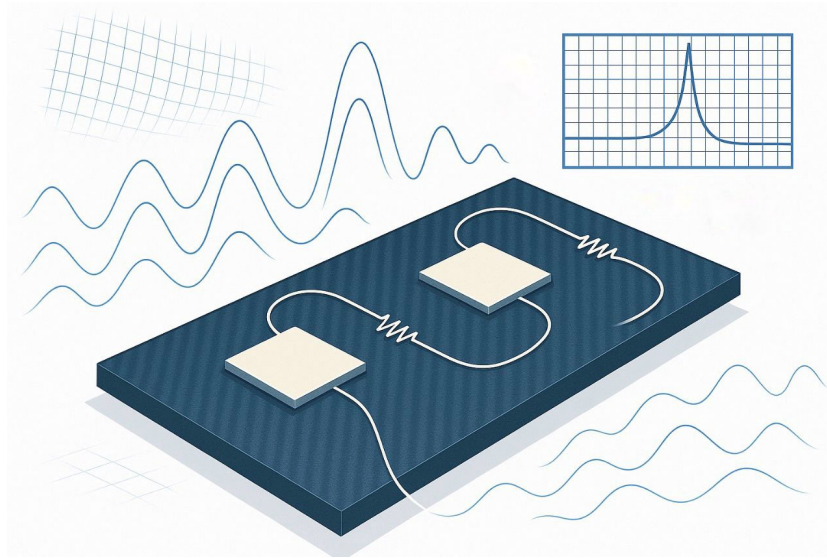




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Multimodal Optimization of Piezoelectric Patch Placement and Shunt Circuit Design for Enhanced Structural Vibration Control

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Thesis to obtain the Master of Science Degree in

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Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

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Resumo

Materiais compósitos têm sido amplamente utilizados em aplicações industriais, substituindo materiais convencionais devido à sua leveza e propriedades mecânicas superiores. No entanto, esses materiais apresentam maior suscetibilidade a vibrações estruturais, o que pode comprometer seu desempenho e vida útil. Para mitigar esse problema, técnicas de controle passivo, como o controle shunt com patches piezoelétricos, têm se destacado por sua simplicidade e eficiência. Essa técnica converte energia mecânica vibratória em energia elétrica, que é então dissipada por meio de circuitos elétricos contendo resistores e indutores. No entanto, desafios importantes ainda persistem, como a definição da posição ótima dos patches e o ajuste dos parâmetros elétricos do circuito.

Nesta tese, foi desenvolvido um modelo numérico em ANSYS para estudar e otimizar a aplicação de patches piezoelétricos em placas compósitas. Foram analisadas duas abordagens de otimização: uma unimodal, que considera apenas o modo de vibração central, e outra multimodal, que leva em conta todos os modos excitados na faixa de frequência de interesse. Os parâmetros foram otimizados utilizando os algoritmos Global and Local Optimization using Direct Search (GLODS) e Direct Multisearch (DMS). Os resultados obtidos permitiram avaliar o desempenho relativo das abordagens e identificar as condições sob as quais cada uma é mais eficaz. Com isso, o estudo fornece diretrizes práticas para a aplicação eficiente do controle shunt em estruturas compósitas submetidas a excitações vibratórias complexas.

Palavras-chave: Materiais piezoelétrico, Controle shunt passivo, Coeficiente de acoplamento eletromecânico, ANSYS, Otimização multimodal.

Abstract

Composite materials have become increasingly prevalent in industrial applications, replacing traditional materials due to their lower structural mass and superior mechanical properties. However, these materials are often more susceptible to structural vibrations, which can degrade performance and reduce operational lifespan. To address this issue, passive control techniques such as piezoelectric shunt damping have gained attention for their simplicity and effectiveness. This method converts vibrational mechanical energy into electrical energy through piezoelectric patches, which is then dissipated via electrical circuits composed of resistors and inductors. Nevertheless, key challenges remain, including the optimal positioning of the patches and tuning of the circuit parameters.

This thesis presents a numerical model developed in ANSYS to optimize the application of piezoelectric patches on composite plates. Two optimization strategies were investigated: a unimodal approach focusing on the dominant vibration mode, and a multimodal approach considering all excited modes within the frequency range of interest. Optimization was carried out using the Global and Local Optimization using Direct Search (GLOS) and Direct Multisearch (DMS) algorithms. The results enabled a comparative assessment of the effectiveness of each control strategy and provided insights into the conditions under which each is most suitable. Therefore, the study offers practical guidelines for the efficient implementation of shunt control in composite structures subjected to complex vibrational excitations.

Keywords: Piezoelectric materials, Passive shunt control, Electromechanical coupling coefficient, ANSYS, Multimodal optimization

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Nomenclature

Greek symbols

α	Geometric ratio between patch length and width
ϵ	Linear strain
ϵ_R	Relative permittivity
γ	Angular strain
μ	Mean of a set of points
Φ	Rotation of the normal to the plate mid-surface
ρ	Density
σ	Mechanical stress

Roman symbols

D	Electric displacement vector
d	Piezoelectric coupling matrix in strain form
E_f	Electric field vector
K	Stiffness matrix
L_d	Differential operator relating displacements to strains
M	Mass matrix
Q_L	Elasticity matrix
S	Flexibility matrix
X	Vector of Cartesian positions of patch pairs
ω	Angular frequency
A_{total}	Total piezoelectric coverage area
E	Euclidean distance of a set of points

f	External non-conservative force
H	Harmonic mean of a set of points
k_e^2	Electromechanical Coupling Coefficient (EMCC)
L	Electrical inductance
l	Length
N_{front}	Total number of points in the Pareto front
$N_{\text{patch pairs}}$	Total number of patch pairs
$N_{\text{variables}}$	Total number of optimization variables
Q	Total modal charge
q	Patch modal charge
R	Electrical resistance
T	Kinetic energy
U	Potential energy
u	Mechanical displacement in the axial direction
W	Work done by non-conservative forces
w	Width
x	Position in the x direction
y	Position in the y direction
Z	Electrical impedance

Subscripts

i, j, k	Computational indices
(m)	M-th vibration mode
F	Force-related properties
$mesh$	Finite element mesh properties
n, n'	Patch pair number
OC	Open circuit
p	Parallel circuit condition
$patch$	Patch-related properties

$plate$	Plate-related properties
ref	Reference value
s	Series circuit condition
SC	Short circuit
u	Mechanical degrees of freedom components
v	Electrical degrees of freedom components
x, y, z	Cartesian components

Superscripts

σ	Constant stress condition
E	Constant electric field condition
T	Transpose

Glossary

ATI	Aerospace Technology Institute
CLPT	Classical Laminated Plate Theory
DMS	Direct MultiSearch
EMCC	Electromechanical Coupling Coefficient
FEM	Finite Element Method
FSDT	First-Order Shear Deformation Theory
GA	Genetic Algorithm
GLDS	Global and Local Optimization using Direct Search
NSGA-II	Non-dominated Sorting Genetic Algorithm II
OP-SW	Opposite Polarity Series Connection
PSO	Particle Swarm Optimization
SP-PW	Same Polarity Parallel Connection
SPAR	Sparse Matrix Solver
TSDT	Third-Order Shear Deformation Theory

Chapter 1

Introduction

1.1 Motivation

Replacement of traditional metallic materials with advanced composites [1] has become increasingly common in automotive, aerospace, robotics, and other high-tech industries. This approach is given because of the unique properties of composites, such as lightweight, flexibility, high strength, and durability, which better meet the specific demands of these sectors, consequently enhancing performance, robustness, and competitiveness of final products, improving both structural, fuel, and cost efficiency while also reducing environmental impacts. In the aerospace industry, the potential of advanced composites was recognized in the 1990s, with predictions that these materials would constitute a significant portion of the structural weight [2]. Based on these early forecasts, recent projections by the Aerospace Technology Institute (ATI) indicate that the global composite market could reach USD 24.8 billion by 2025. This growth is reflected in modern developments, such as Boeing 787 and Airbus A350, where composites now make up 50% of the structural weight and 80% of the structural volume. In the field of robotics, the adoption of composite materials in soft robotics is also important in applications requiring sensitivity, precision, and minimum weight, illustrating the significance of this subject for global advancement.

However, despite the mechanical advantages offered by composites, their adoption also presents structural challenges, where a critical issue associated with these materials arises in high-vibration environments. Under such conditions, composites can exhibit a greater tendency toward structural failures and instabilities, which poses a significant obstacle to their application in contexts such as aircraft fuselage operation and automotive structures subject to significant oscillations. Given this, a potential solution to address this structural dynamic problem is the use of smart materials, as defined in [3], these are structures capable of acting as sensors and taking corrective actions when needed. Examples of these materials include piezoelectric materials, shape memory materials, and magnetorheological materials. Among piezoelectrics, options include ceramic, polymer-based, and composite types, each with specific properties suited for various applications in vibration control.

For the issue at hand, it is especially relevant to focus on piezoelectric materials due to their effectiveness in managing and mitigating structural instabilities. Piezoelectric materials are capable of converting

mechanical energy into electrical energy and vice versa when subjected to a deformation or an electrical potential [4]. Therefore, considering this behavior, the adoption of such smart materials in composite systems to mitigate and absorb vibration energy present at the operating frequency of the structure is an approach increasingly implemented and studied in the literature, leading to the formation of multilayer structures referred to as intelligent structures [5]. However, this approach presents significant challenges when analyzed in depth, both in the electrical, mechanical and geometric domains.

When analyzing the mechanical and geometric domains, the primary focus is on electromechanical coupling, which determines the effectiveness of energy conversion between mechanical and electrical forms in piezoelectric materials. This concept was first introduced as a Electromechanical Coupling Coefficient (EMCC) in [6]. Since then, numerous formulas and methods have been proposed to evaluate the EMCC, with estimations based on a single-mode representation that includes quasi-static effects - known as quasi-static EMCC [7, 8] — or on the effective EMCC, determined both numerically and experimentally [9, 10]. The literature indicates that this coefficient depends on the properties of the piezoelectric patches and their placement on the host structure [11, 12], making the optimal selection of geometric and placement characteristics essential for maximizing the efficiency of vibration energy control. In the electrical domain, it is essential to consider the control types applicable to intelligent structures, these controls can be passive, active or hybrid. Passive control involves applying electrical components, such as resistors and inductors, to dissipate energy [13–15]. Active control, on the other hand, involves implementing a feedback control system that acts as a counter-resonant force [15–17]. Finally, hybrid control combines both feedback mechanisms and electrical dissipation elements [12, 18–20].

Beyond the complexity of the discussed approaches, the modeling and analysis of multilayer structures are also significant challenges. This complexity arises from geometric and material discontinuities, as well as localized electromechanical coupling phenomena. Due to the complexity involved, analytical solutions are only available for a limited number of specific and idealized cases. For this reason, many papers in the literature apply software that employs the Finite Element Method (FEM), such as ANSYS [21], to model these piezoelectric structures. The first implementation of the FEM for solving problems involving piezoelectric structures was developed by Allik and Hughes [22]. Additional insights into this development of FEM for piezoelectric applications are discussed in [23].

As a result, several studies have focused on optimizing the position and geometry of piezoelectric patches to achieve greater coupling between the composite structure and the piezoelectric material, thereby maximizing the EMCC value. Various methods have been proposed in this optimization context, including:

- Parameter Variation Methods, suitable for cases with manageable sample spaces, have been explored by authors such as [24–26].
- Mathematical Programming Methods, particularly Gradient-Based Algorithms, have been highlighted in works as [27–29].
- Stochastic Methods, such as Genetic Algorithms, capable of handling search spaces encompass-

ing both discrete and continuous domains, have been applied by [30, 31].

- Search Methods, algorithms that employ an adaptive grid to find minimum values and are well-suited for black-box problems (where internal equations are inaccessible, such as in ANSYS software), have also been used, such as [13, 32].

Due to the strong problem-solving capability of the latter category, this type of method was chosen for implementation in this thesis. Specifically, the Global and Local Optimization using Direct Search (GLODS) [33] and Direct MultiSearch (DMS) [34] algorithms were employed. These methods have been validated and successfully applied in the literature, as seen in [13, 32].

In addition to optimization techniques for patch placement and geometry, research focused on the calibration of electrical parameters also plays a critical role [13, 15, 32], highlighting the importance of this topic for ongoing study and advancement.

Therefore, considering that composite structures are increasingly relevant in the global market and play a crucial role in advancing various technical sectors, it is essential to adopt measures to mitigate the damage these structures sustain when exposed to significant and potentially unsafe vibrational energy. Accordingly, the scope of this thesis is to develop a numerical model capable of identifying the optimal geometric and electrical characteristics to support the operational demands of the composite material's application under passive control. This thesis aims to create a self-responsive multilayer material, requiring no active actuators, that not only delivers the structural advantages previously outlined but also maintains stability and durability in dynamic application environments.

1.2 Passive Shunt Control

As previously mentioned, there are three main methods for controlling vibration energy in multilayer materials. The first technique explored in the literature is active control, which initially gained prominence due to the impracticality of achieving the high inductance values required for passive control at that time. Although active control was more feasible, it posed significant challenges due to the complex amplifiers and associated sensing electronics necessary for piezoelectric systems. Consequently, hybrid circuits for vibration control were introduced as an alternative solution. However, with technological advancements, high inductance values have been achieved and demonstrated for purely passive inductors using windings around a magnetic coil [35]. This development has made it possible to implement purely passive control using only electrical components such as resistors and inductors, offering a more practical and cost-effective approach to vibration control. To illustrate how these methods can be applied, figure 1.1 shows examples of how piezoelectric materials can use the electrical potential derived from mechanical vibration energy for dissipation and control. In addition, it is important to note that the arrangement shown in the figure is just one of many configurations discussed in the literature, this variability underscores a significant topic of analysis and poses challenges for the proposed problem.

Considering the mentioned techniques, passive control is currently the most relevant area of research and application. Consequently, this thesis will concentrate on passive control as a central theme.

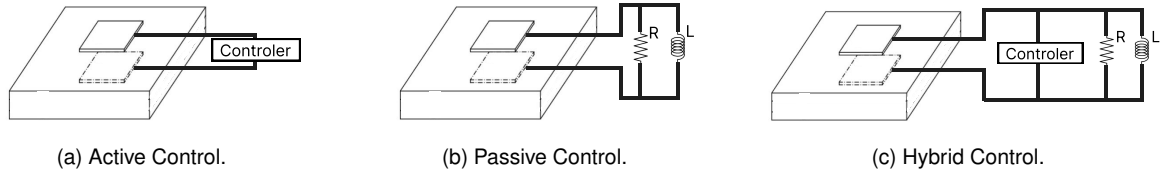


Figure 1.1: General examples of each of the 3 types of control that can be applied to smart structures.

The use of piezoelectric materials connected to resistors and inductors was first explored and experimentally implemented by Forward [15], who demonstrated the influence of resistance and inductance values through experimental analysis of the transducer coupling. In this way, initial calibration methods were introduced by Hagood and Von Flotow [14], where it was applied an RL series shunt to piezoelectric elements and performed calibration using transfer function characteristics and pole placement. Subsequently, Wu [36] developed a calibration method for the RL parallel shunt configuration, which proved to be more practical than the series case. In the parallel configuration, resistance and inductance values are maintained independently, meaning the optimization of inductance does not affect the optimal resistance value. This independence contrasts with the series setup, where both electrical elements share a single dependent potential, impacting their behavior. Since then, several other methods appeared in the literature such as the calibration using residual mode correction introduced first by Krenk and Hogsberg [37] for mechanical absorbers and subsequently extended to piezoelectric shunt damping [38], and a calibration method related with the effective EMCC which was first implemented by Toftekaer [39, 40].

Taking into consideration the complexity and variety of calibration methods for passive control, it is important to emphasize the main classifications and identify which configuration aligns with the scope of this thesis for implementation in the project. The first sort of classification relates to how the equipotential surfaces of the piezoelectric patch pairs are connected. These can be categorized into two types: Same Polarity Parallel Connection (SP-PW) or Opposite Polarity Series Connection (OP-SW). In both cases, the inner surfaces of the patches are connected to the ground. The difference lies in where the electrical circuit terminals are connected. In the first type, SP-PW, the outer surfaces of the patches are kept equipotential, and the electrical circuit is connected between the ground (inner surfaces) and the potential of the outer equipotential surface. In the second type, OP-SW, the electrical circuit is implemented between the outer surfaces of the patch pair. These configurations are illustrated in figure 1.2, where Z is the impedance of the shunt circuit.

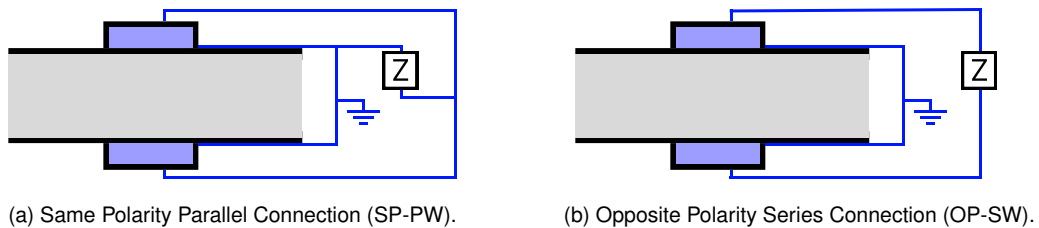


Figure 1.2: Conducting plate structure with a pair of patches wired for a shunt circuit.

Additionally, it is essential to highlight the function of resistors and inductors in the shunt circuit, as both play crucial and distinct roles in the damping effect. By incorporating an inductor in parallel with the capacitive element (i.e., the piezoelectric material), it is possible to tune the inductance to form a resonant circuit. When this inductance is adjusted to match the natural frequency of the structural vibration, it creates an anti-resonant effect. At this frequency, the resistor dissipates the vibration energy, making both components necessary for effective damping. This configuration allows for a classification based on how these electrical elements are arranged. The resistor and inductor can be organized in a series or a parallel model, as shown in figure 1.3. Both configurations achieve structural damping, however, as previously discussed, the parallel model allows for independent tuning and optimization of the inductance and resistance values, simplifying the process [36].

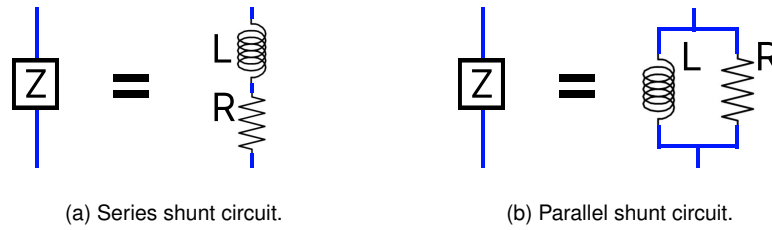


Figure 1.3: Shunt circuit configurations connected to a pair of patches.

Finally, the last type of classification relates to the number of electrical circuits implemented on the structure. This can generally be divided into two types: single shunt circuit or multi-shunt circuit. Both configurations can be effective, but depending on project constraints and scope, one may be more suitable than the other. In the multi-shunt configuration, each pair of piezoelectric patches is connected to its own independent shunt circuit. In contrast, the single shunt configuration involves connecting all patches to a common potential, with a single shunt circuit applied across the terminals of this collective connection, as illustrated in Figure 1.4.

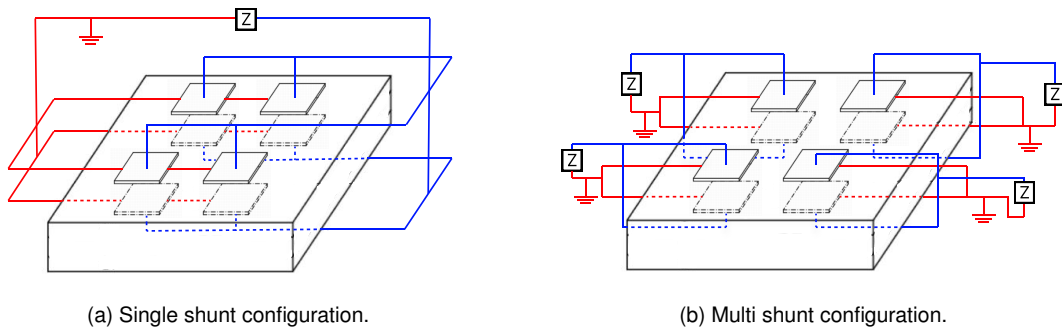


Figure 1.4: Configurations changing the number of shunt circuits connected in the structure considering SP-PW conduction.

Therefore, this brief introduction to passive shunt control demonstrates that the method has numerous branches, making it one of the most studied, analysed, and implemented techniques in both academic literature and industry. Consequently, this thesis will focus on this method. The following sections

will present a detailed discussion of the implementation approach, addressing each of the previously discussed classifications.

1.3 Objectives and Deliverables

The optimization of patch positioning with respect to EMCC aims to assess the influence of the number and placement of patches on the damping effect. Initially, this project focuses on finding the optimal placement for each patch by considering only the effective EMCC for each mode analyzed, both individually and across a frequency range. Once the patches are positioned optimally, the project will then focus on applying passive control techniques, with patches connected in the SP-PW configuration and electrical elements arranged in parallel shunt circuits, choices that will be explained in the following chapters of this thesis. Both single shunt and multi shunt configurations will be examined in scenarios with varying numbers of patches to evaluate their behavior and impact on effective damping, assuming the patches are optimally positioned for the system's frequency range. To achieve this goal, the following logical steps and intermediate objectives are outlined:

1. Research covering the entire project scope, including EMCC calculation methods, numerical models for piezoelectric structures, optimization algorithms from the literature, and RL calibration methods, understanding of existing work and identify unique aspects that can be incorporated into the solution to provide an innovative contribution.
2. Develop a numerical model using FEM in ANSYS that is versatile enough to simulate different positioning, geometric, and electrical conditions, adaptable based on the input provided to the program.
3. Validate and simulate the developed model by comparing it with existing results in the literature for numerical models.
4. Perform the optimization of patch placement using a single-objective approach, where individual modes are considered separately, identifying the optimal modal positions for each number of patches applied to the structure.
5. Optimize patch placement using a multi-objective approach, focusing on modes where their resonance peaks occur near those of other modes.
6. Apply various RL calibration methods and conduct harmonic analysis to evaluate the effectiveness of damping at the resonance peaks for each mode analyzed, considering different numbers of patches.
7. Optimize the values of RL and analyze the harmonic results, comparing these values with those obtained from existing RL calibration methods in the literature.

8. Compare all results obtained using different RL calibration approaches and varying quantities of patches positioned optimally, enabling a conclusion about the influence of patch quantity and RL values on the dissipation of vibration energy within the structure.

1.4 Thesis Outline

This thesis is divided into seven chapters, the contents of which are briefly presented in this section. The current chapter introduces the research topic, discussing the motivation behind the study and explaining the context of its main application: the passive control of vibrations in composite structures through piezoelectric shunt circuits. Additionally, the study objectives and the planned steps to achieve them are outlined. Chapter 2 presents the models used throughout the work that relate the patch and the plate, including theoretical models from the literature, the numerical model developed in ANSYS, and the optimization model applied in the subsequent stages. Chapter 3 describes the implementation of the numerical model using finite elements, detailing all the variables involved, input parameters, and the operation of the program developed in the ANSYS environment. Chapter 4 addresses the implementation of the multi-objective optimization aimed at determining the optimal positioning of the piezoelectric patches on the structure, with an explanation of the objectives considered, the parameters used, and the overall logic of the algorithm. Chapter 5 presents the implementation of the single-objective optimization program, which, based on the previously defined positions, performs harmonic analysis to determine the ideal resistance and inductance values for each shunt circuit. With all parameters established, Chapter 6 discusses the results obtained from both unimodal and multimodal approaches, the latter being proposed and developed in this thesis. Finally, Chapter 7 presents the conclusions of the work, highlighting the main findings, limitations, and suggestions for future research.

1.5 Repository Code

To promote transparency and facilitate the future use of the work presented in this thesis, the following GitHub repository has been created. It contains all the numerical model and optimization codes, along with instructions on how the programs operate and how to use them.

GitHub Repository:

Chapter 2

Piezoelectric-Plate Model

Several complexities arise when analyzing the coupling between plates and piezoelectric patches, making it essential to adopt models that accurately represent the structure's dynamics. Some important topics include the displacement and stress analysis — such as calculating each in both the plate and patch —, the stress, strain and electric field relationships in piezoelectric materials, the equations of motion for the entire structure and the calculation of the Electromechanical Coupling Coefficient (EMCC). These are some of the most relevant aspects that demand careful focus during both the development and analysis stages. In addition to understanding the theoretical piezo-plate relationship, performing FEM analysis is also crucial, such as decisions regarding element selection and mesh design strategies are important areas of study, with extensive research already available in the literature. Finally, it is essential to consider optimization models for patch positioning and geometry, as well as shunt circuit calibration, aiming to provide a brief overview of various optimization approaches used in the literature, with a specific emphasis on the GLODS and DMS models, which are the algorithms applied in this thesis project.

In this chapter, main topics include the theoretical calculations defining the piezo-plate relationship, an overview of the numerical model implemented, and an explanation of the optimization models used, highlighting the main advantages of these methods within the project's context.

2.1 Theoretical Model

As discussed previously, the theoretical model should be developed to understand and illustrate how the coupling relationships between the plate and piezoelectric patch occur. So, the following sections will cover the displacement and stress models, the piezoelectric model relating stress, strain, and electric potential, the model for the structure's equations of motion, and, finally, the calculations of electromechanical coupling, respectively.

2.1.1 Displacements and Strains

The calculation of displacements and strains is the first challenge when analyzing laminated composite structures. Two primary approaches are used: a theoretical two-dimensional (2D) formulation and a numerical three-dimensional (3D) finite element method (FEM).

Two-Dimensional Approach

The theoretical formulation for displacement in laminated structures is a well-explored topic in the literature, mainly, when the problem is reduced from a three-dimensional one to a two-dimensional. Theoretical models include the Classical Laminated Plate Theory (CLPT), First-Order Shear Deformation Theory (FSDT) and Third-Order Shear Deformation Theory (TSDT), as can be seen in [41–43]. The primary difference among these methods lies in the order of the expansion used to define displacement.

Given that the plate and patch dimensions have much smaller thicknesses compared to other dimensions, transverse displacement is assumed to be constant, leading to the following general expressions for displacement:

$$\begin{aligned} u(x, y, z, t) &= u_0(x, y, t) + z\Phi_x(x, y, t) - c_1 z^3 \left(\Phi_x + \frac{\partial w_0}{\partial x} \right) \\ v(x, y, z, t) &= v_0(x, y, t) + z\Phi_y(x, y, t) - c_1 z^3 \left(\Phi_y + \frac{\partial w_0}{\partial y} \right) \\ w(x, y, z, t) &= w_0(x, y, t) \end{aligned} \quad (2.1)$$

where c_1 is a constant that specifies which of the previously discussed models is being implemented. If $c_1 = 0$, the equation represents the CLPT and FSDT models, as they are of first order. Conversely, if $c_1 \neq 0$, the model is of third order, representing the TSDT model.

So, using Equation (2.1) to determine displacement in each interval and knowing that the relationship between strain and displacement is given by $\epsilon = \frac{du}{dx}$, the following strain equations are defined:

$$\begin{aligned} \epsilon_{xx} &= \epsilon_{xx}^{(0)} + z\epsilon_{xx}^{(1)} + z^3\epsilon_{xx}^{(2)} \\ \epsilon_{yy} &= \epsilon_{yy}^{(0)} + z\epsilon_{yy}^{(1)} + z^3\epsilon_{yy}^{(2)} \\ \epsilon_{zz} &= 0 \\ \gamma_{yz} &= \gamma_{yz}^{(0)} + z^2\gamma_{yz}^{(2)} \\ \gamma_{xz} &= \gamma_{xz}^{(0)} + z^2\gamma_{xz}^{(2)} \end{aligned} \quad (2.2)$$

where ϵ is the linear strain and γ is the angular strain, more details are defined in [44].

Given these equations, the next step is to define the relationship between strain and stress for both the plate and the piezoelectric patch, which has a distinct relationship.

Three-Dimensional Approach

Nevertheless, as previously stated, ANSYS was used to solve the problem numerically, diverging from the two-dimensional approach illustrated before and instead adopting a three-dimensional methodology. To achieve this, the software implements a three-dimensional finite element approach based on

solid mechanics principles to calculate displacement and strain fields. This method accounts for material properties, boundary conditions, and applied loads, thereby ensuring an accurate representation of the plate and piezoelectric patch behaviour.

The FEM software manages the relationship between displacement and strain through its finite element formulations, discretizing the geometry into elements and solving the equations of motion numerically. For laminated composites, it incorporates anisotropic material properties, capturing the specific deformation mechanics of each layer.

This numerical approach provides a practical and efficient solution for complex geometries and loading conditions, complementing the theoretical formulations discussed earlier.

2.1.2 Stress formulation and relation

The calculation of stress in relation to strain, when defined in principal coordinates for an orthotropic material — materials with unique and independent properties in the three directions — which is the case for the plate and piezoelectric patches used in the problem addressed in this thesis, can be defined by the following expression:

$$\sigma = Q_L \epsilon \quad (2.3)$$

where Q_L is the elasticity matrix.

However, although this expression may be a good representation in the case of the plate, for piezoelectric patches, the attempt must be changed due to the stress, strain and electric field relationships that piezoelectric materials present.

Piezoelectric materials possess the characteristic of converting mechanical energy into electrical energy, and vice versa. When a piezoelectric material deforms, it generates an electric potential, or when a specific electric potential is applied to the material, it induces deformation [6]. Therefore, piezoelectric materials are described by equations that relate strain, stress, and electric field. These equations can be written as follows [45]:

$$\begin{aligned} \epsilon &= S^E \sigma + d^T E_f \\ D &= d \sigma + \epsilon^\sigma E_f \end{aligned} \quad (2.4)$$

where σ is the mechanical stress vector, ϵ is the mechanical strain vector, D is the electric displacement vector, E_f is the electric field vector, S^E is the compliance matrix, d is the piezoelectric coupling matrix in strain form and ϵ^σ is the dielectric matrix measured at constant stress.

2.1.3 Equations of Motion

The equations of motion can be defined using the d'Alembert Principle, which states that the sum of the kinetic energy, potential energy, and the work done by non-conservative forces must be equal to zero, this principle is commonly used in the analysis of dynamic systems with dissipative forces, such as damping or friction. This can be mathematically represented as follows:

$$\Delta T - \Delta U + \Delta W = 0 \quad (2.5)$$

where T represents the kinetic energy, U represents the potential energy, and W the total work done on the system by external non-conservative sources (mechanical and electrical).

Accordingly, each energy term can be described as follows:

$$\partial T = - \int_{\Omega} \partial (u^T \rho \ddot{u}) d\Omega \quad (2.6a)$$

$$\partial U = \int_{\Omega} \partial (\epsilon \sigma - E_f D) d\Omega \quad (2.6b)$$

$$\partial W = \int_{\Omega} \partial (u^T f + v^T Q) d\Omega \quad (2.6c)$$

As can be seen above, the only different terms are in the potential energy, $E_f D$, which corresponds to the electric potential energy within the piezoelectric material and in the non-conservative work term, $v^T Q$, where Q denotes the electric charge. Thus, Equation (2.6b) can be further developed by applying the following well-known relationships: $\epsilon = L_d u$, where L_d is a differential operator, $E_f = -\nabla v$ and the relationships on strain and stress in both the mechanical and electrical domains. In this way, the potential energy equation can be defined as follows:

$$\partial U = \int_{\Omega} \left[\partial (u^T L_d^T Q_L^{E_f} L_d u) - \partial (u^T L_d^T e \nabla v) - \partial (v \nabla^T e^T L_d u) - \partial (v \nabla^T E_f^e \nabla v) \right] d\Omega \quad (2.7)$$

With all terms now defined, these expressions can be substituted into the main Equation (2.5), yielding the following expression:

$$\int_{\Omega} \left\{ \partial \left[u^T (\rho \ddot{u} + L_d^T Q_L^{E_f} L_d u + L_d^T e \nabla v - f) \right] + \partial \left[v^T (\nabla^T e^T L_d u - \nabla^T E_f^e \nabla v + Q) \right] \right\} d\Omega = 0 \quad (2.8)$$

Therefore, this formula allows to define the equation of motion of the proposed system. However, as can be seen, the solution turns out not to be so trivial, so it is necessary to adopt the finite element model for the solution, this is done by making the variables u and v actually interpolation functions, which will define the nodes of the finite element system. Consequently, by rearranging the Equation (2.8) and expressing it in matrix format, the following expression is obtained:

$$\begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \ddot{u} \\ \ddot{v} \end{Bmatrix} + \begin{bmatrix} \mathbf{K}_{uu} & \mathbf{K}_{uv} \\ \mathbf{K}_{uv}^T & \mathbf{K}_{vv} \end{bmatrix} \begin{Bmatrix} u \\ v \end{Bmatrix} = \begin{Bmatrix} \mathbf{f} \\ -\mathbf{Q} \end{Bmatrix} \quad (2.9)$$

where $\mathbf{K}_{uu}$, $\mathbf{K}_{uv}$ and $\mathbf{K}_{vv}$ are the stiffness matrices associated with the mechanical displacements, electromechanical coupling and electric potentials, respectively, $\mathbf{M}$ contains the physical mass associated with the inertia of the plate and piezoelectric patches, $\mathbf{Q}$ is the applied electric charges vector, ω is the circular frequency, u is the vector of mechanical degrees of freedom and v is the vector of electric potentials.

2.1.4 Electromechanical Coupling Coefficient (EMCC)

Having defined the equation of motion that governs the system, it is now essential to formulate the equations representing the equilibrium conditions for free vibrations, taking into account the piezoelectric effect. This expression can be derived from Equation (2.9) and is given as follows:

$$\left[\begin{bmatrix} \mathbf{K}_{uu} & \mathbf{K}_{uv} \\ \mathbf{K}_{uv}^T & \mathbf{K}_{vv} \end{bmatrix} - \omega^2 \begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \right] \begin{Bmatrix} u \\ v \end{Bmatrix} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{Q} \end{Bmatrix} \quad (2.10)$$

A precise definition of the free vibration equation is fundamental to ascertain the extent to which the piezoelectric material exerts an influence on the behaviour of the structure, quantified by the EMCC. As indicated in section 1.2, a number of techniques for calculating this coefficient have been proposed in the literature. However, as this thesis project was designed to be solved using FEM software, the chosen calculation method will focus on the effective EMCC. By solving Equation (2.10), it is possible to determine the eigenvalue, and, although this approach is computationally demanding, it yields more accurate results by avoiding approximations, providing a clearer and more detailed understanding of the structure's dynamics. So, the effective coefficient can be calculated as follows:

$$k_{e(m)}^2 = \frac{\omega_{OC(m)}^2 - \omega_{SC(m)}^2}{\omega_{SC(m)}^2} \quad (2.11)$$

where $\omega_{OC(m)}$ represents the natural frequency of mode m of the system under the open-circuit condition, characterized by an electrical impedance $Z = \infty$, which implies that the electric charge $\mathbf{Q}$ is zero in Equation (2.10). Similarly, $\omega_{SC(m)}$ corresponds to the natural frequency under the short-circuit condition, in which $Z = 0$, indicating that $\mathbf{Q} \neq 0$ in Equation (2.10). It is worth noting that the EMCC value is usually presented as a percentage.

2.1.5 Calibration of Electrical Elements

As discussed in Section 1.2, several methods exist for calibrating resistance and inductance to attenuate vibration peaks in piezoelectric structures. In this thesis, the approach based on EMCC, as detailed in the previous section, was adopted. Among the RL calibration models implemented, the one based on the tuning procedure from [40] is highlighted. The resulting expressions for parallel circuits (subscript p) are given below:

$$R_s = \frac{k_{e(m)}^2 \omega_{SC}}{Q_{SC(m)}^2} \sqrt{\frac{2k_{e(m)}^2}{(1 + k_{e(m)}^2)^3}} \quad L_s = \frac{k_{e(m)}^2}{Q_{SC(m)}^2 (1 + k_{e(m)}^2)^2} \quad (2.12)$$

$$R_p = \frac{k_{e(m)}^2 \omega_{SC(m)}}{Q_{SC(m)}^2} \sqrt{\frac{1}{2k_{e(m)}^2}} \quad L_p = \frac{k_{e(m)}^2}{Q_{SC(m)}^2} \quad (2.13)$$

where $Q_{SC(m)}$ and $\omega_{SC(m)}$ are the modal charge and short-circuit natural frequency, respectively, computed by Equation (2.10) for mode m .

2.2 Numerical Model

The implementation of FEM has become increasingly prevalent as a tool for analyzing and solving piezoelectric models, as demonstrated in [23]. To address and analyze the problem proposed in this thesis, a numerical model was developed and implemented using the ANSYS software [21].

To achieve the objectives of the project, the code developed in the software was designed to be sufficiently flexible, allowing the simulation of different geometric and electrical configurations, as well as modal and harmonic analyses of the piezo-plate structure. This flexibility aimed to optimize the positioning of piezoelectric patch pairs on the plate and calibrate the electrical values, representing a crucial step in validating the proposed project, where is a key part of the motivation for the thesis scope.

To clarify the rationale and methodology implemented in the ANSYS code, the following sections, along with the flowchart in Figure 2.1, provide a step-by-step explanation of each phase of the process. These can be divided into three main parts: Model Generation, which is a standard step in the process; Modal Analysis, which involves calculating natural frequencies, modal charges, and the EMCC; and Harmonic Analysis, where the effective vibration reduction is evaluated.

ANSYS Mechanical APDL

Model Generation

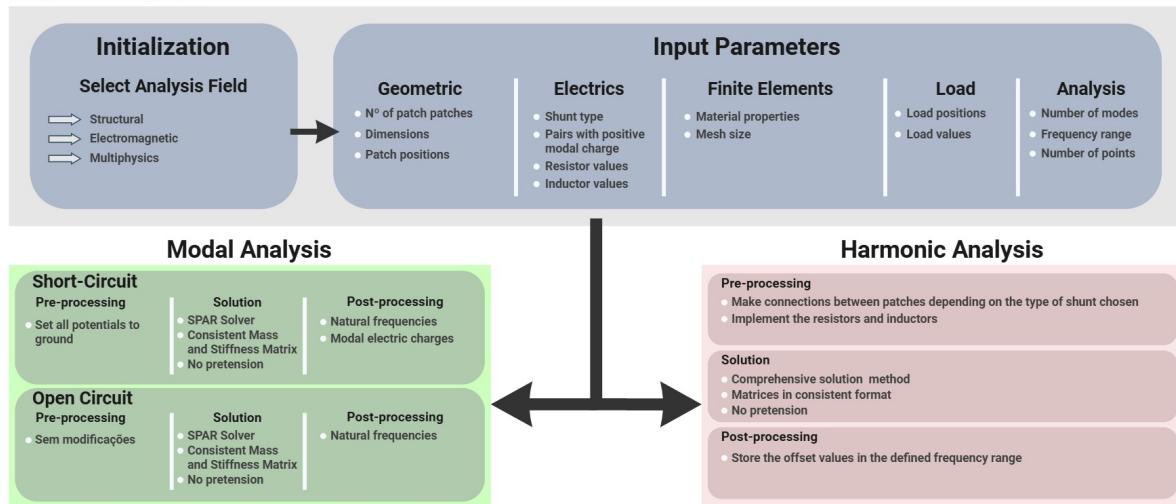


Figure 2.1: Flowchart of the ANSYS program implemented in the thesis.

2.2.1 Model Generation

The Model Generation step corresponds to the pre-processing phase of the finite element method. The primary goal in this stage is to construct the structural model and apply boundary conditions. For the implementation, this was achieved through the following workflow:

1. ANSYS Startup

The initial stage of the ANSYS configuration process is to define the requisite fields of analysis to address the problem's scope, which encompasses both mechanical and electrical domains. To

achieve this, the simulation is configured to include structural analysis for mechanical properties, electromagnetic analysis for electrical behavior, and multi-physics analysis to capture the coupling between these domains.

2. Input parameters

For the numerical model to achieve sufficient flexibility, it is essential to allow the simulation of various geometric and electrical configurations. The input parameters define key aspects of the model, which can be grouped into the following categories:

- **Geometric Parameters:** These parameters include the number of patches and the dimensions of the plate and piezoelectric patches, such as length, width, and thickness. Additionally, the position of each patch pair on the plate is specified by defining the x and y coordinates for each pair.
- **Electrical Parameters:** These parameters define the type of shunt circuit used (single or multiple) along with the associated electrical components, such as resistors and inductors. It's important to note that when a single shunt circuit is applied, only one value for each component (resistor and inductor) is used, regardless of the number of patches. In contrast, for a multiple shunt circuit, different values for resistors and inductors can be applied to each pair. These settings are essential for evaluating various electrical configurations, particularly in harmonic analyses.
- **Finite Element Parameters:** This category defines the mesh sizes for both the plate and the patches, ensuring proper discretization for accurate simulation results. Material properties are also specified here, including the relevant parameters for both the plate and the patches.
- **Load Parameters:** These include the magnitude and location of applied forces (x and y coordinates), which are crucial for simulating the harmonic behavior of the structure.
- **Analysis Parameters:** These determine the scope of the analysis, including modal and harmonic evaluations, and allow for the investigation of resonance frequencies and vibration reduction effectiveness.

The flexibility of these input parameters ensures the model can simulate a range of configurations and conditions. The table 2.1 provides a detailed account of the parameters included in each of the aforementioned categories.

3. Finite element model generation

Once the parameters defined, the next step in the Model Generation process is the development of the finite element model. This involves creating the geometric model, defining and discretizing the mesh, establishing contact surfaces between the patches and the plate, and specifying the equipotential surfaces. Each of these steps is described in detail below.

Table 2.1: Input parameters for the numerical model in ANSYS.

Category	Parameter	Dimension	Description
Geometric	Number of patch pairs	Integer	Number of patch pairs.
	Plate Length	Real value [m]	Defines the length of the plate.
	Plate Width	Real value [m]	Defines the width of the plate.
	Plate Thickness	Real value [m]	Defines the thickness of the plate.
	Patch Length	Real value [m]	Defines the length of the patch.
	Patch Width	Real value [m]	Defines the width of the patch.
	Patch Thickness	Real value [m]	Defines the thickness of the patch.
	Patch pair position	Real values [x,y] [m]	Specifies the coordinates x and y of each patch pair.
Electrical	Shunt Type	Categorical	{0: Multiple, 1: Single}
	Pair with positive modal charge	Boolean	Defines if the patch pair has a positive modal charge
	Resistor	Real value [Ω]	Real values representing resistor values for each circuit.
	Inductors	Real value [H]	Real values representing inductor values for each circuit.
	Circuit Type	Categorical	{1: Short-Circuit, 2: Open-Circuit, 3: Only Resistor, 4: RL circuit in Series, 5: RL circuit in Parallel}
Finite Element	Material Properties	Real values	Material properties for the plate and patches (e.g., Young's Modulus, Poisson's ratio).
	Mesh size	Real value [m]	Specifies the size of the mesh.
Load	Load positions	Real values [x,y] [m]	Positions x and y where force is applied.
	Load value	Real value [N]	Real values representing the magnitude of the applied force.
Analysis	Modal Analysis		
	Number of modes	Integer	Specifies the number of modes to be calculated.
	Harmonic Analysis		
	Minimum Frequency	Real value [Hz]	Specifies the minimum frequency for the analysis.
	Maximum Frequency	Real value [Hz]	Specifies the maximum frequency for the analysis.
	Number of points	Integer	Defines the number of frequency points to be calculated.

(a) Geometric Model

The first step of the model generation is create the geometric model. This process leverages for loop functions in ANSYS to create the model efficiently. Using the predefined patch positions and the dimensions of both the patches and the plate, the code systematically generates keypoints, lines, areas, and volumes required for the simulation. Figure 2.2 demonstrates this process for a generic case with four patch pairs, highlighting how the automation ensures consistency and accuracy in model generation.

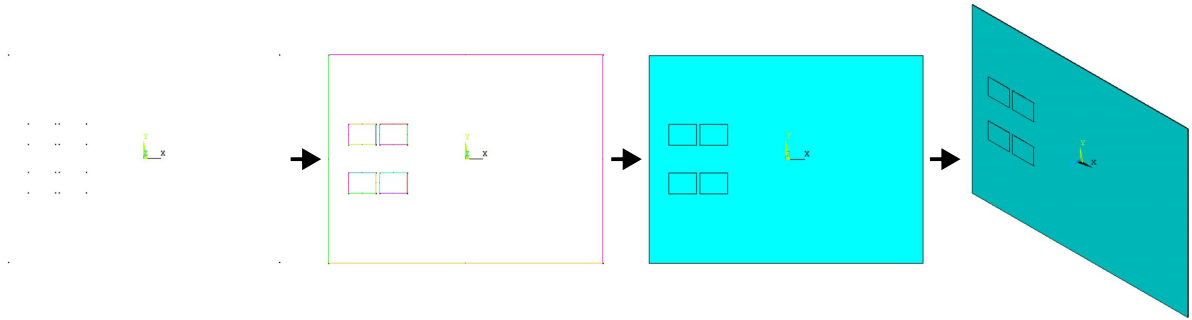


Figure 2.2: Process of automatic geometric model creation in ANSYS for a 4 patch pairs model, showcasing keypoints, lines, areas, and volumes, respectively.

This step was crucial to the progress of the research, as the ability to automatically simulate various configurations enables the use of optimization algorithms for each case. This approach ensures that the most effective geometric configuration is efficiently identified for any proposed problem.

(b) Finite Element Model

The next step in model generation involves creating the mesh domain and discretizing the geometry. This requires careful consideration of parameters related to material properties and element types. It is important to emphasize, as previously mentioned, that a 3D approach was adopted for this project, consequently, element types such as SOLID are suitable for this model, more details about the choices made will be provided in chapter 3.1. Additionally, since the problem involves relatively simple structures, a structured mesh was adopted to enhance the numerical solution's speed and efficiency. An example of this is illustrated in figure 2.3, which a structured mesh is applied in a plate structure with four patch pairs.

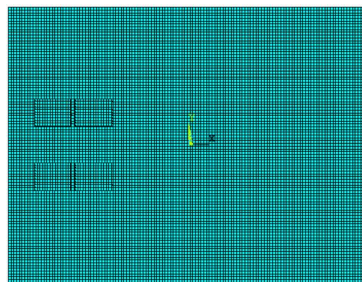


Figure 2.3: Structured Mesh profile for a 4 patch pairs model.

(c) **Contact Surfaces**

The definition of contact surfaces is essential to accurately represent the interaction between the plate and the piezoelectric patches. In practical applications, adhesive materials are used to bond the patches to the plate, ensuring strong adhesion and eliminating any possibility of separation. However, to simplify the numerical model and reduce computational complexity, this study assumes a no-separation condition without explicitly modeling the adhesive layer. This assumption provides a balance between realistic behavior and computational efficiency, while still ensuring an accurate representation of the structural interaction.

In order to achieve this condition, a node-surface contact and a no separation condition were applied. This condition ensures that there is no gap or relative movement at the interface during the simulations, effectively preventing any separation between the patch and the plate, which maintains the structural integrity of the assembly.

(d) **Equipotential Surfaces**

Finally, the last step in the model generation is the definition of the equipotential surfaces for the patches. As explained in Section 1.2, there are two methods for connecting the patch surfaces: Same Polarity Parallel Connection (SP-PW) and Opposite Polarity Series Connection (OP-SW). For this project, the SP-PW configuration was selected. In this arrangement, since the patch pairs are positioned symmetrically, when the plate vibrates, the patches can combine modal charges effectively, producing a significant electric voltage. In contrast, in the OP-SW configuration, the patch positions result in an approximately zero potential difference across the impedance.

So, in the SP-PW configuration, for each patch pair, the outer surfaces of both patches are equipotential, and the inner surfaces of each patch within the pair are also equipotential. Consequently, the electric impedance is defined between the potential difference of the outer and inner equipotential surfaces, as illustrated in Figure 1.2a.

4. **Boundary conditions**

The boundary conditions for the model are divided into two domains: Electrical and Mechanical.

In the electrical domain, the boundary condition applied is a grounding condition. For each pair of patches, the inner surfaces of both patches are electrically connected to the ground.

In the mechanical domain, the boundary conditions ensure that the plate is properly constrained. The model assumes simply supported conditions for the plate, where the displacement in the vertical direction (Z-axis) is restricted at all edges. This means that the plate cannot move out of the plane. Additionally, two corners of the plate are fully restrained to ensure that the problem is fully constrained. This is achieved by applying zero displacement conditions in both the X and Y directions at the corners, which prevents any translational movement in these directions.

Once the previously outlined workflow has been correctly applied, the piezoelectric-plate structure is effectively established, incorporating a degree of flexibility in its parameters. This structure is now ready

for both modal and harmonic analyses. However, for each of these analyses, additional preprocessing steps related to electrical circuit configurations still need to be implemented. Despite this, the foundation of the numerical model is fully established at this stage.

2.2.2 Modal Analysis

With the finite element model of the piezoelectric-plate structure already defined, the next step is the Modal Analysis, which involves solving the problem under two distinct conditions: short-circuit and open circuit. The resonance frequencies for both conditions are stored, and using these values, the effective EMCC is calculated as given by Equation (2.11). The workflow for each condition is detailed below:

- **Open Circuit**

1. Pre-processing

The open-circuit condition is defined such that the electric terminals of the patches are disconnected. As already outlined in section 2.2.1, no additional modifications are required in the preprocessing stage for the open-circuit condition.

2. Solution

In the solution phase, modal analysis begins after completing the preprocessing steps. The analysis type is set to modal, and key parameters are defined, including the *Number of modes* - present in table 2.1 - to be analyzed and the solution method, which employs the efficient *Sparse Matrix Solver (SPAR)*. The mass and stiffness matrices are represented in their consistent (non-lumped) form, ensuring a precise distribution of mass and stiffness across the structure. Pre-stress effects are disregarded in this analysis, as they are not relevant to the current configuration. This setup allows the solver to accurately and efficiently compute the system's natural frequencies and mode shapes.

3. Post-processing

The postprocessing phase focuses on extracting the natural frequencies for each analyzed mode, which are essential for calculating the EMCC.

- **Short-Circuit**

1. Pre-processing

The concept of a short-circuit refers to an electrical circuit condition where the impedance is effectively zero. Based on this principle, since the inner surfaces of the patch pairs are already defined as grounded, the only adjustment required is to set the outer surfaces of the patch pairs to the same ground potential. This ensures that both terminals are at the same potential, effectively creating a short-circuit condition.

2. Solution

The solution phase is the same already defined previously in Open Circuit configuration.

3. Post-processing

After completing the solution phase, postprocessing focuses on extracting the key results: the natural frequencies and modal charges for each analyzed mode. The natural frequencies also for calculating the EMCC, while the modal charges represent the system's "reaction forces" under excitation. This last result is mainly significant in the single-shunt approach, where the electrical connections between patches with the same modal charge polarity must be considered to ensure accurate representation and effective design for a vibration reduction.

In summary, the expected outcomes of the modal analysis include obtaining information such as the natural frequencies, the Electromechanical Coupling Coefficient (EMCC), and the modal charges.

2.2.3 Harmonic Analysis

In addition to modal analysis, harmonic analysis is a critical aspect of the project as it evaluates the effectiveness of vibration reduction when resistors and inductances are implemented in the piezoelectric patch pairs. This analysis assesses the system's dynamic response to harmonic excitations, providing insights into its performance under operational conditions. To carry out this step, it is necessary to define the electrical configurations of the system. The process can be divided into the following steps:

1. Pre-processing

The first step in the harmonic analysis involves defining the electrical configuration. There are two possible shunt configurations: single and multi, as illustrated in figure 1.4.

In a single shunt circuit, all patches are connected, resulting in a single value for both resistance and inductance. It is important to emphasise that in this type of circuit, the modal charges of the patches exert a significant influence on the connections between them. In the typical case where all patches have positive modal charges, the outer surfaces of the patches are connected together, and the inner surfaces are connected in a similar manner. However, if any patch exhibits a negative modal charge, the connection strategy is altered. In such cases, the patch with the negative modal charge requires alternating connections: its inner surface is connected to the outer surfaces of the other patches, and its outer surface is connected to the inner surfaces of the other patches. This information about the modal charge is communicated to the numerical model through the parameter *Pair with positive modal charge*, as defined in table 2.1.

On the other hand, in a multi shunt circuit, each patch pair is connected to its own dedicated electrical circuit. Both configurations were implemented and compared within the scope of this project to evaluate their performance.

After selecting the type of shunt configuration, the arrangement of the resistor and inductor can be defined, as present in figure 1.3. For this analysis, it is important to emphasize that the parallel shunt approach was adopted.

In addition to defining the electrical configuration, another crucial preprocessing step in harmonic analysis is the application of loads to the structure. Parameters such as the load magnitude and its

position are specified in Table 2.1, ensuring that the simulation accurately represents the intended operating conditions.

2. Solution

A comprehensive solution method is employed to ensure precise results across all degrees of freedom, particularly important for capturing complex dynamic behaviors. The matrix representation is considered in its consistent (non-lumped) form, reflecting the system's detailed characteristics. Pre-stress effects are excluded in this analysis to simplify the computations and focus on the harmonic response. The frequency range of interest is specified, along with the number of frequency points for analysis, providing a clear resolution of the system's response across the selected spectrum. Additionally, the analysis uses stepwise loading to accurately capture the behavior at each defined frequency step. This configuration ensures a robust evaluation of the system's harmonic performance.

3. Post-processing

Once the solver completes its execution, the postprocessing phase focuses on evaluating the effectiveness of vibration reduction in the structure. This is achieved by analyzing the amplitude at the point of force application, enabling an assessment of the performance of the passive control mechanism and its impact on minimizing structural vibrations under harmonic excitation.

In summary, the expected outcome of the harmonic analysis is the determination of the displacement amplitude across a defined frequency range, offering insights into the system's vibrational behavior under harmonic excitation with an applied passive control.

2.3 Optimization Model

Following the implementation of the numerical model, the subsequent stage involves the selection of an optimisation algorithm to achieve the desired objectives. As outlined briefly in section 1.1, the central focus of this thesis is the resolution of problems characterised by a multitude of possibilities, which will be addressed through the utilisation of black-box algorithms. In this instance, the ANSYS software will be employed as the computational tool. In light of this, the project employs search methodologies, given their capacity to effectively explore extensive solution spaces. Among these methods, Direct MultiSearch (DMS) algorithm was selected for its robust exploration capabilities and suitability for the problem at hand.

2.3.1 Direct Multisearch Algorithm

The DMS multi-objective optimization algorithm, as detailed in [34], was incorporated into this study to address the necessity for a multi-objective approach. This approach was adopted once it was determined that the problems to be solved consist mainly in optimising two or more objectives. For example, the number of patches will always be an objective to be optimised within a given range of frequency. If

this range is defined for just one mode, this constitutes the study of a unimodal function, but still in a multi-objective approach, since the number of patches and the chosen mode will be the focus.

The method implemented in the DMS program is designed to identify the Pareto front in multi-objective optimization problems. The Pareto front represents the set of optimal solutions for the specific problem, comprising all non-dominated points. Each point on this front signifies a solution in which improving any one objective would lead to a deterioration in at least one other objective. Unlike single-objective approaches, the DMS algorithm produces a set of non-dominated solutions, each ensuring that improving one objective inevitably leads to a reduction in at least one other.

2.3.2 Global and Local Optimization Using Direct Search

Unlike DMS, the GLODS optimization algorithm, as presented in [33], was implemented in this project with the objective of solving single-objective problems. In the context of this thesis, this problem refers to determining the optimal values of the electrical components based on the reduction of the vibration peak observed in the harmonic response.

Given this, GLODS is particularly well-suited for global optimization tasks that are derivative-free and subject to bound constraints. Developed with the specific goal of identifying as many local minima as possible, GLODS employs a directional direct search procedure, which makes it efficient in identifying the global minimum of the evaluated function.

2.3.3 Problem Variables

The variables that the optimization program needs to optimize consist of 2 different approaches:

Optimization of Patch Pair Positions

The first optimization problem focuses on determining the optimal positions of each patch pair on the plate, in alignment with the objectives of the study. It is essential to note that each patch pair is associated with a single coordinate (x, y) , resulting in two positional variables per pair. This means that for any given patch pair, the top and bottom patches are always aligned at the same position. Consequently, if the problem involves optimizing the positions of 4 patch pairs, a total of 8 positional variables must be optimized. This relationship can be expressed more generally by the Equation (2.14).

$$N_{\text{variables}} = 2 \cdot N_{\text{patch pairs}} \quad (2.14)$$

Optimization of Electrical Components

The second approach focuses on optimizing the resistance and inductance values. As described in Section 1.2, the electrical arrangement can be configured in two ways: single shunt or multi shunt. In the single shunt configuration, the number of variables is always 2, corresponding to one resistor and one inductor. Conversely, in the multi shunt configuration, the number of electrical component variables

is double the number of patch pairs, as each pair has its own circuit with one resistor and one inductor. This relationship is expressed in equation (2.15).

$$\begin{aligned} \text{Single Shunt} &\implies N_{\text{variables}} = 2 \\ \text{Multi Shunt} &\implies N_{\text{variables}} = 2 \cdot N_{\text{patch pairs}} \end{aligned} \tag{2.15}$$

2.3.4 Objective Functions

As with the variables, the objective functions differ depending on the two optimization approaches:

Optimization of Patch Pair Positions

The objective of the position optimization is to maximize the effective EMCC values and the number of patch pairs. As discussed in Section 2.1.4, the EMCC coefficient captures the influence of the patch pair distribution on the plate's behavior. This optimization aims to enhance the EMCC for each patch pair configuration by adjusting the positional variables for each pair.

To achieve this, a mathematical workaround was necessary. Since the DMS optimization algorithm is inherently designed to minimize a function, the approach involved minimizing the negative of the EMCC and the number of patch pairs. This method ensures that the desired maximum is obtained as the negative of the computed minimum:

$$\max(F(x)) = -\min(-F(x)) \tag{2.16}$$

where $F(x)$ represents the vector of objectives, which, in this case, includes the values of k_e^2 (EMCC) for each mode analyzed, and the last element corresponding to the number of patch pairs.

Optimization of Electrical Components

In this approach, the objective function is subject to change, since the patch pairs are already positioned in their optimal configuration as a result of the previous optimization step. At this stage, the goal is to minimize the vibration peak in the harmonic problem under analysis. The vibration peak is evaluated by selecting a specific point on the plate and assessing the deformation experienced at that point, which varies depending on the values of resistance and inductance applied—these being the variables of the problem.

To obtain the optimal results, the GLODS algorithm was implemented with the aim of minimizing the following function:

$$F(x) = \max(\mathbf{D}(\mathbf{x})) \tag{2.17}$$

where $F(x)$ represents the maximum displacement value of the variable $\mathbf{D}(\mathbf{x})$, which corresponds to the harmonic response curve of the displacement in the z -direction over the analyzed frequency range, considering the electrical configuration defined by $\mathbf{x}$.

2.3.5 Patch Pair Distribution

Optimization problems involving placement are a significant area of interest and research in the literature, particularly because many of these problems operate within the real domain. This means that the range of possible solutions is theoretically infinite, posing a considerable computational challenge. This issue is also present in the problem discussed in the scope of this thesis, where the placement of each patch pair exists within the continuous domain equivalent to the plate size.

To address this challenge, a common numerical technique is to implement a discretization of the main domain, thereby reducing the infinite set of possibilities into a manageable, finite set of candidate solutions. In addition to this challenge, placement optimization also faces the issue of overlap, where, depending on the number of structures considered, the proportion of non-physical solutions (i.e., solutions with overlapping patches) may increase, creating additional obstacles for optimization algorithms.

This thesis addresses these two challenges using two numerical strategies, which are explained in detail in the following sections.

Percentage Position

To transform the continuous domain into a discrete domain, the first step involves converting the positional intervals into a percentage-based representation. The intervals for the placement of each patch depend on the dimensions of the plate and the patch itself, as the patch pair cannot extend beyond the edges or be positioned too close to the edges, ensuring the entire patch pair remains within the plate area. The generalized expression for these intervals can be defined as:

$$\begin{aligned} x_n &\in \left[-\frac{l_{plate}}{2} + \frac{l_{patch}}{2} + l_{mesh} ; \frac{l_{plate}}{2} - \frac{l_{patch}}{2} - l_{mesh} \right] \\ y_n &\in \left[-\frac{w_{plate}}{2} + \frac{w_{patch}}{2} + l_{mesh} ; \frac{w_{plate}}{2} - \frac{w_{patch}}{2} - l_{mesh} \right] \\ n &= 1, 2, \dots, N_{\text{patch pairs}} \end{aligned} \quad (2.18)$$

where, l_{plate} represents the plate's total length, and w_{plate} is its total width. The variables l_{patch} and w_{patch} correspond to the length and width of the individual patches, respectively. The parameter l_{mesh} refers to the mesh size, which ensures that the patches are positioned with sufficient clearance from the plate edges to maintain numerical stability and prevent boundary-related issues. The indices $n = 1, 2, \dots, N_{\text{patch pairs}}$ denote the number of patch pairs being considered. This formulation ensures that the patches are constrained within a safe, feasible area on the plate.

By converting this nominal interval into a percentage-based approach, the values of x_n and y_n are normalized to vary between 0 and 1, creating a dimensionless domain.

This percentage-based transformation simplifies the implementation of discretization. Once normalized, the interval becomes more general and independent of the number of patch pairs, enabling the optimization process to be more efficient. For example, applying a discrete domain with an increment of 0.1 converts the continuous domain into a manageable subset, represented as a grid of discrete points. Figure 2.4 illustrates the transformation between nominal and percentage-based intervals in a case with two patch pairs applied to the plate structure.

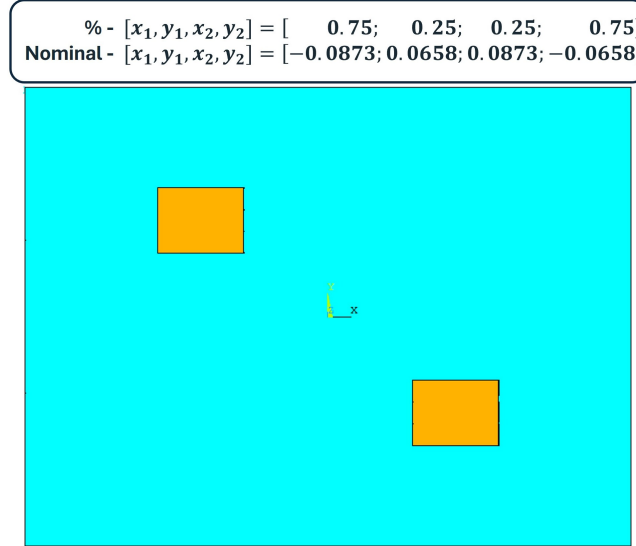


Figure 2.4: Position transformation between nominal and percentage-based intervals.

Using this technique in the optimization program provides several advantages. The program can start with a rude discretization (a larger increment), allowing for faster initial optimization. The discretization can then be progressively refined, either until the solution stabilizes (i.e., no significant changes in the results occur) or until a desired precision is achieved.

Another key improvement of this method is its compatibility with the DMS algorithm. The algorithm can begin with the optimal solutions from the rude discretization and use them as starting points for subsequent refinements. This stepwise refinement significantly reduces computational time, even as the number of possible solutions increases with finer discretization, enabling the program to efficiently converge on the best solution.

Overlap Prevention

To prevent overlap cases, a strategy was implemented to evaluate the distribution of patch pairs prior to running the ANSYS simulations. This process is handled by a MATLAB script designed to verify the distribution of patch pairs proposed by the optimization algorithm.

The workflow begins with the optimization algorithm generating a potential solution. The script then verifies the number of patch pairs in the proposed solution and performs a sorting operation. This sorting step arranges the patch center coordinates in descending order of their x -axis and y -values, respectively. By standardizing the order of patch pairs, the program avoids redundant evaluations of solutions that are mathematically identical but differ in the order of patch pairs, thereby saving computational time.

Next, the algorithm checks for overlaps between patch pairs. To ensure numerical stability in ANSYS, a minimum distance is enforced between the centers of the patch pairs. This distance is defined as the patch size plus one mesh element size in both horizontal and vertical directions. In other words, the inequality shown in Equation (2.19) must be satisfied. If any overlaps are detected, the script adjusts the distribution by removing the overlapping patch pairs, thereby reducing the total number of pairs in the solution. This step ensures that the resulting configuration is both feasible and compliant with the

optimization constraints.

$$\begin{aligned}
|x_n - x_{n'}| &\geq l + l_{\text{mesh}} \\
|y_n - y_{n'}| &\geq w + l_{\text{mesh}} \\
n &= 1, \dots, N_{\text{patch pairs}} - 1 \\
n' &= n + 1, \dots, N_{\text{patch pairs}}
\end{aligned} \tag{2.19}$$

Figure 2.5 illustrates an example of this algorithm. As shown, the program resolves overlap cases by transforming an infeasible solution into a valid one, ultimately reducing the number of patch pairs in the final configuration.

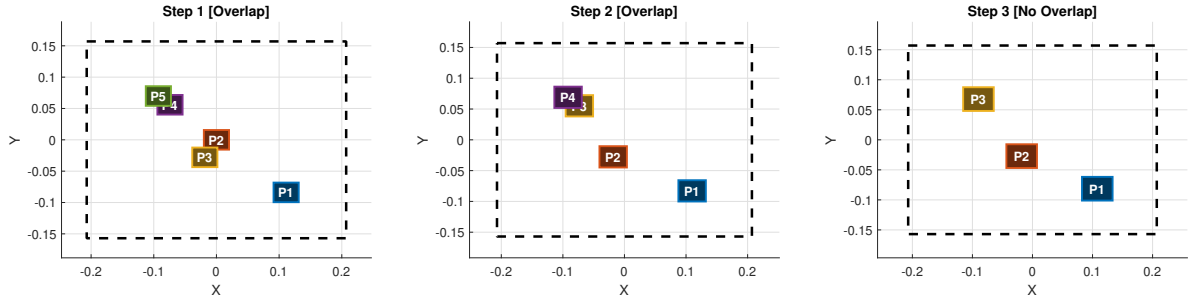


Figure 2.5: Example of the overlap prevention algorithm, starting with an initial percentage distribution of 5 patch pairs given by the vector $zz = [0.25 \ 0.75 \ 0.30 \ 0.70 \ 0.50 \ 0.50 \ 0.45 \ 0.40 \ 0.80 \ 0.20]$.

2.3.6 Patch Pair Dimensions

One of the key assumptions in the scope of this thesis is to maintain the total mass of piezoelectric material constant, regardless of the number of patch pairs. This assumption ensures that comparisons between different quantities of patch pairs are exclusively dependent on the division of the material into more pairs, rather than an increase in the total mass. Without this assumption, increasing the number of patch pairs would always result in better performance compared to using a single patch, as the covered area would be larger. By maintaining the mass (or area) constant, this issue is avoided, and the results depend solely on the number of patch pairs.

To achieve this, the total area of piezoelectric material was fixed as $A_{\text{total}} = 0.0052 \text{ m}^2$. Additionally, a standard geometry ratio for the patch pairs was defined, fixing the ratio between length and width as $\alpha = 1.3185$. With these standard values established, the geometric parameters of each patch pair can be determined based solely on the number of patches applied in the problem. The relationships used to calculate these values are defined by the following equations:

$$w_{\text{patch}} = \sqrt{\frac{A_{\text{total}}}{\alpha \cdot N_{\text{patch pairs}}}} \tag{2.20a}$$

$$l_{\text{patch}} = \alpha \cdot w_{\text{patch}} \tag{2.20b}$$

where l_{patch} and w_{patch} represent the geometric dimensions of each patch pair, and $N_{\text{patch pairs}}$ denotes the total number of patch pairs.

Chapter 3

Implementation

Once all the models were defined, this chapter presents a detailed description of the implementation of each program. First, the ANSYS implementation is discussed, including the main input parameters required to run the simulations and the expected outputs for each case. Then, the optimization programs are described in more detail, explaining how various challenges of the problem were addressed and how the programs were executed to achieve the best results within an optimal computational time and with efficient use of computational resources.

3.1 ANSYS Implementation

With the workflow of the numerical model already outlined in Section 2.2, this section focuses on its implementation in ANSYS. The objective here is to detail the process, emphasizing material properties, element selection, dimensions, and all essential parameters required for the simulation setup.

The section is structured as follows: first, it addresses the fixed parameters and model setup, i.e., the parameters that remain constant throughout the entire thesis problem. Next, it discusses the variable parameters of the model, providing guidance on how users can incorporate these parameters into the model, along with practical examples. Finally, the chapter concludes with a verification process, comparing the results of the numerical model with existing literature. This comparison ensures the validity and reliability of the model for addressing the problems within the scope of this thesis.

3.1.1 Fixed Parameters and Model Setup

Regarding the input parameters shown in table 2.1, the parameters that remain constant are primarily related to the plate dimensions and patch thickness. Parameters associated with the finite element definitions, such as the materials and elements representing the patch and plate structures, were kept consistent throughout all analyses within the scope of this thesis. Similarly, the load value was also maintained constant, ensuring uniformity in the applied conditions. The model analyses were conducted up to 10 vibrational modes, and the number of harmonic analysis points was predefined and remained

unchanged throughout all simulations. Table 3.1 summarizes these fixed parameter values, along with descriptions about each choice.

Table 3.1: Fixed parameters for the numerical model in ANSYS.

Category	Parameter	Value	Description
Geometric	Plate Length	0.414 [m]	
	Plate Width	0.314 [m]	
	Plate Thickness	0.001 [m]	
	Patch Thickness	0.0005 [m]	
Finite Element	Plate Material	Aluminum	Material properties described in Table 3.3
	Plate Element	SOLID186	Structural element for modeling solid materials in 3D, handling mechanical stress and deformations.
	Patch Material	Anisotropic Piezoelectric Ceramics	Material properties described in Table 3.2
	Patch Element	SOLID226	Structural element where both mechanical and electrical behaviors are important and need to be coupled in the analysis.
	Circuit Element	CIRCU94	Circuit element for use in piezoelectric-circuit, it's applicable to full harmonic and transient analyses.
	Mesh Size	3.175×10^{-3} [m]	The mesh size is smaller than the element thickness.
Load	Load Value	-100 [N]	The load value is fixed at 100 N in the downward direction.
Analysis	Modal Analysis		
	Number of Modes	10	The modal analysis was limited to 10 vibrational modes.
	Harmonic Analysis		
	Number of Points	162	This number was defined based on comparisons with numerical results and provides satisfactory accuracy.

Table 3.2: PZT-5H properties.

$S^E [\times 10^{12} Pa^{-1}]$	$d [\times 10^{-12} C/N]$	ϵ_R	$\rho [kg/m^3]$
$S_{11}^E = 16.5$	$d_{31} = -274$	$(\epsilon_R)_{11} = 3130$	7500
$S_{12}^E = -4.78$	$d_{31} = -274$	$(\epsilon_R)_{22} = 3130$	
$S_{13}^E = -8.45$	$d_{33} = 593$	$(\epsilon_R)_{33} = 3400$	
$S_{33}^E = 20.7$	$d_{24} = 741$		
$S_{44}^E = 43.5$	$d_{15} = 741$		
$S_{55}^E = 43.5$			
$S_{66}^E = 42.6$			

Table 3.3: Aluminum properties.

$E [\times 10^9 N/m^2]$	ν	$\rho [kg/m^3]$
70	0.33	2700

3.1.2 Model Variables and Input Parameters

The parameters that vary for each analysis can be divided into two main groups. The first group consists of variables related to the optimization process, as described in Section 2.3. These include the number of patch pairs, their positions, and resistor and inductance values. Although patch dimensions are not optimization variables, they are included in this group because their values are calculated based on the number of patch pairs, as outlined in Section 2.3.6. The second group consists of parameters that the user must manually define at the beginning, depending on the type of analysis that wish to perform. These include additional parameters related to the electrical circuits or other analysis-specific configurations.

Each of these two groups is detailed further, along with instructions on how the user can provide these input parameters to the main ANSYS program.

1. Optimization Variables

These variables are set externally by the optimization program. To facilitate this integration, input values are provided through text files that the program reads to execute the analysis. Specifically, three main text files must be prepared: *Input.GeometricParameters*, *Input.PositionPatches*, and *Input.ElectricValues*.

The *Input.GeometricParameters* file contains geometric information, including the number of patch pairs and their dimensions. The *Input.PositionPatches* file specifies the nominal coordinates for each patch pair. The expected format for the first file includes three values, while the second file requires $2 \cdot N_{\text{patch pairs}}$ values, representing the x and y coordinates of each patch pair.

Lastly, the *Input.ElectricValues* file defines the type of electric circuit implemented in the shunt (e.g., single or multi-shunt) and the corresponding resistor and inductor values. The size of this file depends on the selected shunt type. For a single shunt, the file is expected to contain three values: the circuit type, the resistor value, and the inductor value. For a multi-shunt configuration, the file size scales with the number of patch pairs. It must include $3 \cdot N_{\text{patch pairs}}$ values: the first $N_{\text{patch pairs}}$ numbers represent the resistor values, followed by $N_{\text{patch pairs}}$ inductor values, finally, the last $N_{\text{patch pairs}}$ values specify the circuit type for each patch pair. Figure 2 provides examples of these three input files.

2. User-Defined Parameters

These variables are configured directly in the ANSYS script and are defined by the user based on the specific problem that needs to be solved. These parameters include those related to the electrical properties, load settings, and analysis models.

The shunt type, as described in section 1.2, determines the number of circuits present in the system. The ANSYS program is designed to automatically adapt to different configurations based on the shunt type selected by the user. The user simply specifies the desired configuration to match their problem requirements. Additionally, the program is generalized to handle cases with

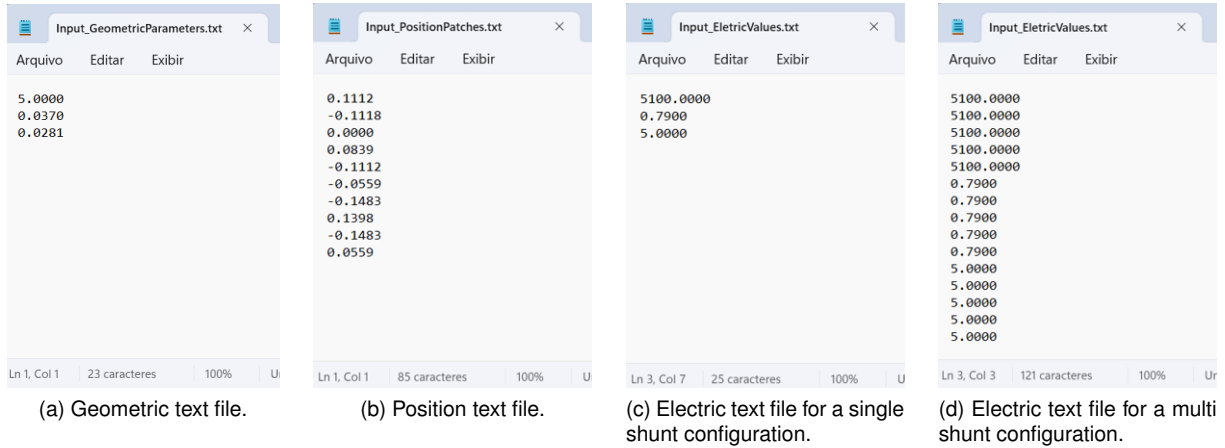


Figure 3.1: Example of input text files for a configuration with 5 patch pairs.

a single patch pair, where the connections between patch pairs must be defined based on modal charges. The user can specify which patch pairs should be connected for their analysis.

The circuit type for each patch pair also needs to be defined in this section. This parameter is important for both modal and harmonic analyses. In the modal analysis, the circuit types related to short-circuit and open-circuit conditions are set automatically to calculate the Effective EMCC. In the harmonic analysis, the user can select the specific circuit type to implement.

The load position has also been generalized, allowing the user to define any number of loads at any position, enabling a more detailed evaluation of their influence on the harmonic analysis. Finally, the harmonic analysis setup has been made flexible to accommodate the desired frequency range specified by the user, providing the ability to tailor the analysis to the problem's needs.

3.1.3 Model Verification

The numerical model developed in this thesis is validated against results presented in the literature. Verification is performed through both modal and harmonic analyses. For modal verification, natural frequencies are compared in both open-circuit and short-circuit conditions, thus validating the results related to the Effective EMCC and other variables. In the harmonic analysis, validation is carried out by comparing the resonant peaks and its harmonic behavior in frequency response.

In both verification processes, the configuration shown in figure 3.2 was implemented, featuring a system with four patch pairs, each positioned at the center of its respective quadrant. The following sections provide a detailed comparison and analysis of the verification results.

Modal Verification

For modal verification, the same conditions and problem setup described in [40] were implemented, using it as the reference for comparison. Table 3.4 presents a detailed comparison between the modal results obtained from the model developed in this thesis and those reported in the reference study.

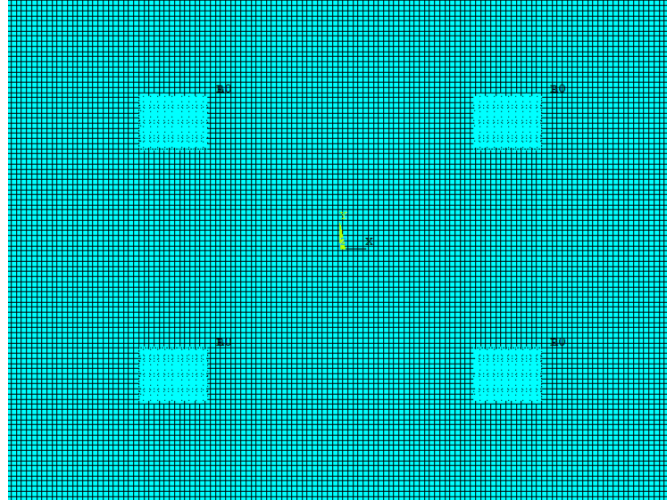


Figure 3.2: Plate-piezoelectric configuration used to verify the thesis model.

Table 3.4: Comparison of modal results for the first 8 vibrational modes between the Thesis Model and the Reference Model for a case with 4 patch pairs.

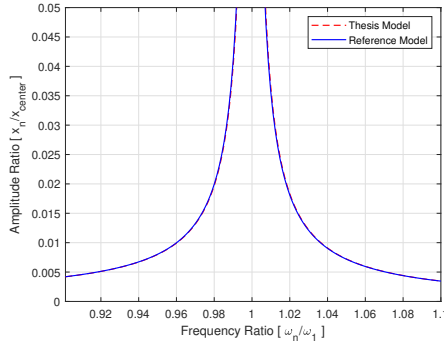
Mode	Thesis Model					Reference Model [40]					EMCC Error
	Short-Circuit		Open-Circuit			Short-Circuit		Open-Circuit			$\Delta k_e^2/k_{e,r}^2$
	f_{SC} [Hz]	Q_r $10^3[C]$	f_{OC} [Hz]	V_r $10^{-4}[V]$	k_e^2 [%]	f_{SC} [Hz]	Q_r $10^3[C]$	f_{OC} [Hz]	V_r $10^{-4}[V]$	$k_{e,r}^2$ [%]	
1	38.58	2.89	38.66	2.10	0.41	38.58	2.88	38.66	2.10	0.41	0.00
2	78.25	9.22	78.66	6.71	1.02	78.24	9.17	78.64	6.69	1.01	0.01
3	109.4	12.4	109.9	8.99	0.94	109.3	12.3	109.8	8.98	0.93	0.01
4	143.4	23.1	144.7	16.9	1.93	143.3	22.9	144.7	16.9	1.90	0.02
5	153.0	14.2	153.5	10.3	0.63	153.0	14.1	153.5	10.3	0.62	0.02
6	219.9	25.4	221.0	18.6	0.99	219.8	25.2	220.9	18.4	0.98	0.01
7	236.3	19.9	236.9	14.5	0.52	236.3	19.8	236.9	14.4	0.52	0.00
8	254.6	0.00	254.6	0.03	0.00	254.6	0.00	254.6	0.01	0.00	0.00

As shown in table 3.4, the EMCC error values are negligible, indicating that the thesis model is capable of performing modal analyses with high precision and without significant errors. This demonstrates a reliable verification, confirming that the model is suitable for achieving the modal objectives of this thesis.

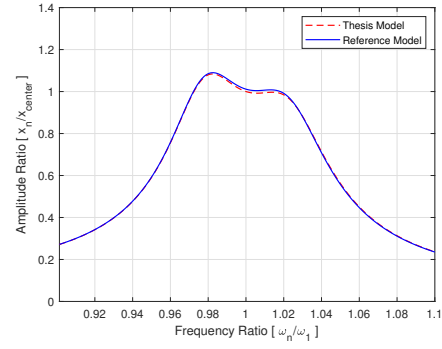
Harmonic Verification

For the harmonic verification, the reference paper [46] was used. The objective of this comparison is to assess the model's accuracy by analyzing the frequency response of the first vibrational mode in a system with four patch pairs. Figure 2 presents the frequency response curves for both the Thesis Model and the Reference Model, based on the circuit configurations specified in the reference study.

To quantitatively compare the two curves in Figure 2, metrics such as the Residual Sum of Squares (RSS) were calculated. The resulting RSS value was $6.16 \cdot 10^{-4}$, indicating that both curves exhibit highly similar behavior. This confirms that the thesis model is capable of accurately solving harmonic problems.



(a) Frequency response without passive control.



(b) Frequency response in a multishunt circuit with $R = 332.8 \text{ k}\Omega$ and $L = 122.8 \text{ H}$.

Figure 3.3: Frequency response for the first vibrational mode without control application and with control application, respectively.

3.2 DMS Implementation

With the numerical model established, the focus now shifts to optimization. The first program to be discussed is the Direct MultiSearch. After defining and describing the DMS program in Section 2.3, the discussion now moves to its implementation, illustrating how the algorithm is applied to obtain optimal results for each approach outlined in this thesis.

First, this section discusses the startup parameters integrated into the program and their application within the analysis flow. Additionally, the chapter details the workflow of the implemented DMS program, including variables related to discretization and how the program manages the search for different numbers of patch pairs. Finally, the chapter concludes by verifying both problem approaches—optimal position and optimal RL values — to ensure that the program effectively addresses the thesis problem.

3.2.1 Algorithm

The DMS program algorithm can be divided into two main parts: a general framework that remains constant across different optimization problems, and a specific implementation designed to achieve the optimization objective.

General Algorithm

The foundation of the method relies on the principles of direct search applied to multi-objective problems, structured around maintaining a list of non-dominated points (L_k). At each iteration, the method selects an active point from this list and performs a local search consisting of two stages: search and poll.

In the search stage, a finite set of new candidates is generated around the selected point. These candidates are evaluated and compared with the existing list, retaining only non-dominated solutions that effectively improve some objective. If new points are found in this stage, the iteration is considered successful and no further action is required. Otherwise, the poll stage is triggered, which systematically

evaluates directions from a positive spanning set to ensure complete local coverage. The update of the step parameter α_k regulates the balance between exploration and refinement: in successful iterations, the step is maintained or enlarged, whereas in unsuccessful ones, it is reduced, allowing a progressive approach to the Pareto front.

A fundamental difference compared to classical population-based methods, such as GA, PSO, and NSGA-II, is that DMS is a point-based algorithm. This means that the search evolution is guided by individual points and their associated vector space, rather than relying on the simultaneous evolution of an entire population of solutions. This characteristic reduces the need for massive problem evaluations, ensuring that exploration occurs in a structured and incremental way, thereby covering the search space without the typical overhead of population-based methods.

In the present project, the initial configuration of DMS is defined in the `parameters_dms.m` file, where the most critical parameter is `list`, which determines the initial sample for the optimization process. This sample can be generated randomly or retrieved from previously stored data.

To ensure an efficient search for optimal solutions, the initialization process follows a structured approach. Initially, the program starts with a coarser discretization (larger increments) and uses a randomly generated sample set, allowing a broad search of the solution space. After this preliminary exploration, the discretization is refined, gradually increasing precision. At this stage, the `list` parameter is modified to use stored samples from previous runs, concentrating the search in the most promising regions. This strategy helps balance global exploration with local refinement, enabling the algorithm to converge efficiently to Pareto solutions without excessive computational overhead.

Specific Algorithm - Optimization of Patch Pair Positions

One of the challenges in implementing the DMS algorithm for problems involving varying numbers of patch pairs is handling the dynamic number of optimization variables. As described in Equation (2.14), the number of variables depends on the number of patch pairs considered in the problem.

To address this challenge, we define a maximum allowable number of patch pairs for optimization, thereby maintaining a fixed number of optimization variables. However, to accommodate different patch pair configurations, a special handling mechanism is introduced: if any coordinate variable (x, y) is assigned a value of -1 , the algorithm interprets this as an inactive patch pair. In other words, if at least one coordinate of a patch pair is set to -1 , that patch pair is excluded from the optimization process.

This approach ensures flexibility while maintaining a structured optimization framework. Figure 3.4 illustrates how the program manages different patch pair configurations using this method. The process consists of the following four steps:

1. **Reshaping the Input Variable (Step I):** The input variable, generated by the DMS program, is reshaped into a matrix where the first column corresponds to the x -coordinates and the second column to the y -coordinates.
2. **Handling Inactive Patch Pairs (Step II):** The program searches for any occurrence of -1 . If found, it sets all coordinate values (x, y) for that patch pair to -1 , marking it as inactive.

3. **Reordering the Matrix (Step III):** The program reorders the matrix in descending order of x -coordinates and, in case of ties, by descending order of y -coordinates. This step prevents redundant computations by ensuring that solutions differing only in position ordering are not considered separately.
4. **Reshaping and Output Validation (Step IV):** The matrix is reshaped back to its original format. The output variable is then checked for overlapping patch pairs. If no overlap is detected, the ANSYS program evaluates the objective function, considering only the active patch pairs.

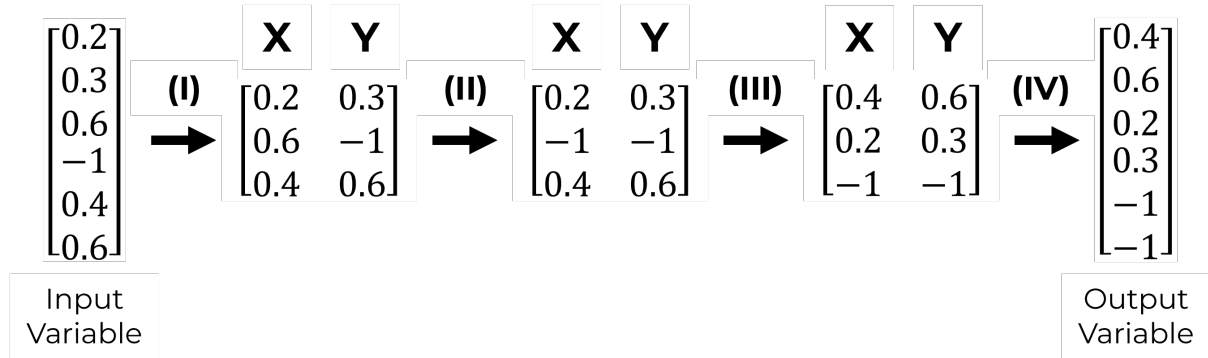


Figure 3.4: Procedure for managing input variables to optimize different numbers of patch pairs.

3.2.2 Initial Analysis of the Model

To verify whether the optimization model is achieving the expected results, preliminary analyses are necessary. Since DMS is responsible for finding the optimal solution to the problem involving the distribution of patch pairs, model validation is performed by analyzing this distribution, considering specific modes in which the optimal positions are trivially defined based on the modal analysis of the structure's displacement. In addition, the results are compared with those found in the existing literature.

Thus, considering the first vibration mode, where the amplitude peak occurs exclusively at the center of the plate, it is expected that DMS will provide, as an optimal solution, the placement of the patches in this central region—regardless of the number of patch pairs used. To verify this, the model was tested using up to 5 patch pairs and a maximum percentage discretization of 0.01. Figure 3.5 illustrates the distribution of patch pairs for this first vibration mode.

As illustrated in Figure 3.5, the DMS program positioned the patch pairs at the center of the plate, as expected for the first vibration mode, in which the vibration peak occurs in this region. Furthermore, it is important to emphasize that although Figures 3.5b, 3.5c, and 3.5d show maximum deformation at the edges of the plate (red color present in the figures), this discrepancy is solely due to the reference in z used by ANSYS for the analysis. Thus, the vibration peak remains at the center of the plate, regardless of the reference adopted in the solution. This result demonstrates that the optimization model is capable of handling this type of problem with high accuracy, as discretizations as fine as 0.01 were implemented and validated.

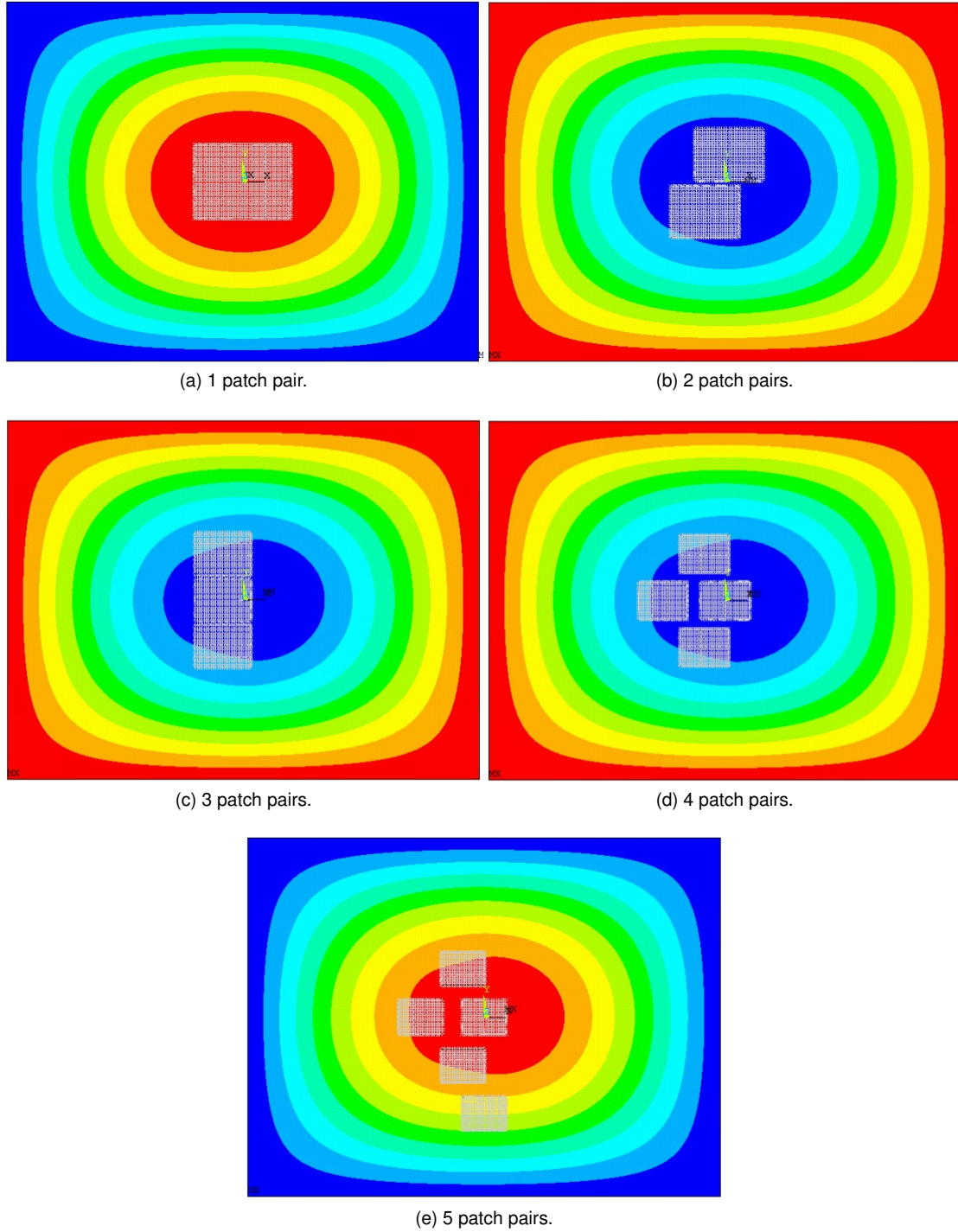


Figure 3.5: Model validation of the optimization for the first vibration mode using up to 5 patch pairs applied to the plate.

3.3 GLODS Implementation

Finally, the implementation of the GLODS program is more straightforward compared to the DMS algorithm presented earlier. Applied to the optimization of electrical components, GLODS operates without the need for explicit constraints or filtering of infeasible solutions, as discussed in Section 3.2.1.

In this case, the problem variables—resistance and inductance—are treated freely, being limited only by the intervals defined for their admissible values.

This section details the implementation flow of the program, highlighting the main parameters used and how they contribute to obtaining optimal solutions.

3.3.1 Algorithm

GLODS can be understood as a model very similar to the direct search method employed in DMS, but adapted to the context of global single-objective problems. While DMS evolves a list of non-dominated points, GLODS conducts multiple parallel local searches starting from well-distributed initial points in the search space. The alternating search/poll structure is preserved, but with important differences:

- In the search stage, different global sampling strategies (random sampling, Latin Hypercube, Sobol, Halton, 2^n -Centers) allow broad exploration of the domain;
- In the poll stage, each local search systematically refines promising regions;
- Parallel searches converging to nearby regions are merged, ensuring that the number of active processes corresponds to the number of distinct local minima.

This combination gives GLODS a hybrid character between multistart and direct search, reducing redundancies and increasing efficiency in identifying the global minimum.

Accordingly, the program initialization parameters are defined in `parameters_glods_progressivo.m`, with emphasis on the variables `list` and `nPini`. The variable `list` determines whether the initial sampling points are generated randomly or loaded from previously saved data. The parameter `nPini` defines the number of initial points with which the optimization will begin.

Thus, the implementation follows the execution flow below:

1. **Define a fixed geometric and positional structure for the patches:** At this stage, the geometric characteristics of the model are already optimal. Therefore, the GLODS optimization program operates in a scenario where the patch positions on the plate are fixed.
2. **Define a coarse discretization:** Initially, a less refined discretization grid is adopted for resistance (R) and inductance (L), with the goal of identifying promising intervals in the search space and avoiding excessive computational cost in less relevant regions.
3. **Initialize with random points:** With the parameter limits defined, the algorithm is started with points randomly distributed within these limits, ensuring good initial coverage of the search space. This process is defined by the parameter `list`.
4. **Define a more refined discretization:** After the preliminary analysis with coarse discretization, the most promising regions are refined. This allows the algorithm to more precisely explore the resistance and inductance values that showed better initial performance.

5. **Initialize with the best points already stored:** To leverage the results of the previous stage and improve computational efficiency, the best points found are used as the basis for a new optimization round with the refined grid. This process is controlled by the parameter `list`. This approach guides the search toward regions with greater improvement potential, accelerating convergence to the optimal solution.

3.3.2 Initial Analysis of the Model

To assess whether the optimization model converges to coherent results, a preliminary analysis procedure was defined. In this process, the position of the piezoelectric patch was fixed in advance, and based on this configuration, the harmonic response was used to verify whether the model is capable of identifying suitable values for resistance and inductance.

Since this is an initial analysis involving a heuristic model, there is no exact metric to quantify the correctness of the values found. However, it is possible to evaluate whether the algorithm converges to solutions that are consistent with the expected physical behavior of the system.

A simple configuration was adopted, with only one patch operating over a frequency range corresponding to the first vibration mode of the structure. The variable ranges were defined as $R \in [42, 47] \text{ k}\Omega$ and $L \in [3, 6] \text{ H}$, with discretizations of $100 \text{ }\Omega$ and 0.1 H , respectively. After running the program, the optimal values obtained were $R = 46 \text{ k}\Omega$ and $L = 36.5 \text{ H}$. Figure 3.6 shows a comparison of the harmonic response of the structure with and without the optimized passive control.

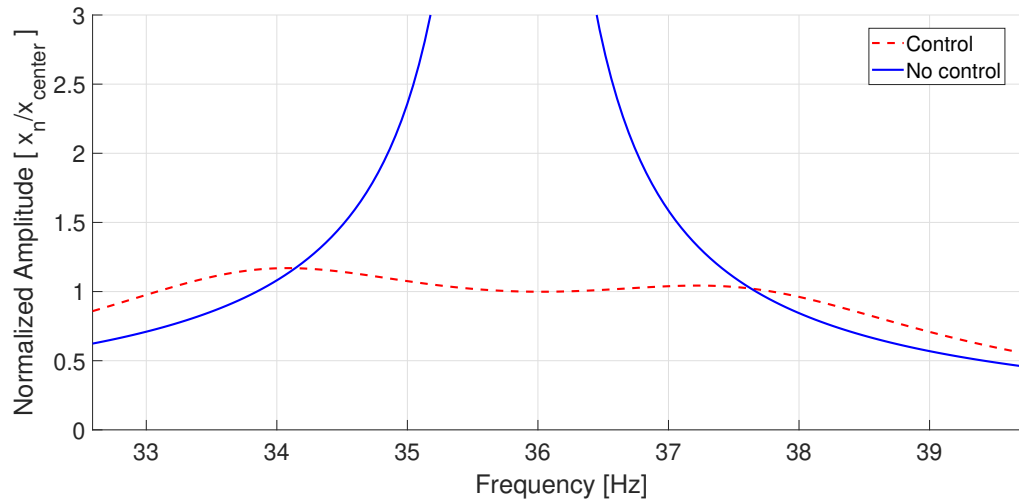


Figure 3.6: Initial analysis of the resistance and inductance optimization model based on the maximum amplitude within the analyzed frequency range.

It can be observed that the program was able to identify appropriate values for R and L , achieving tuning with the peak of the first vibration mode and its attenuation, confirming that the algorithm is effective for this type of problem. It is worth noting that, as described in the execution flow, the results can be further refined with denser discretizations — which were not applied here, since the goal was only to validate the overall functioning of the model.

Chapter 4

Results and Discussion

This chapter presents and discusses the results obtained through modal and harmonic analyses based on the developed numerical model. Initially, a unimodal approach is adopted, in which the first ten natural modes of the structure are analyzed individually. For each mode, the optimal configuration of up to five pairs of piezoelectric patches is sought, using the EMCC value as the performance criterion. These geometric configurations are then evaluated through harmonic analysis, using passive single shunt circuits with RL electrical parameters calibrated according to the analytical expressions presented in Equation 2.13. The objective is to assess the effectiveness of vibration peak reduction in both low and high-frequency modes.

Based on the results of this stage, a multimodal approach is proposed, aimed at mitigating vibrations in high-frequency modes. In this case, the optimization simultaneously considers a set of modes to define the best patch positions, followed by calibration of the electrical parameters using optimization techniques. Finally, a comparison is made between the different methods analyzed, highlighting the advantages and limitations of each approach, as well as their most suitable applications in the context of passive vibration control in smart structures.

4.1 Unimodal Approach

4.1.1 Problem Description

As discussed throughout this thesis, vibration mitigation in smart structures is a multifaceted problem that requires distinct modeling and control approaches. In this context, the unimodal strategy proposes an analysis based on the isolated consideration of each vibration mode, making it suitable for applications where the excitation is concentrated in a narrow frequency range, encompassing only one mode at a time.

This approach consists of determining the optimal positions and quantities of piezoelectric patch pairs for the first ten natural modes of the structure, aiming to maximize the electromechanical coupling measured by the index $k_{e(m)}^2$. Based on the best results obtained, the RL electrical parameters of the passive shunt circuits were calculated using Equation 2.13, assuming a parallel circuit configuration.

From these configurations, harmonic analysis was carried out to verify the effectiveness of passive control in attenuating the vibratory response.

Thus, the unimodal problem can be organized into the following steps:

1. Modal Analysis

- (a) Consideration of the first 10 natural modes of the structure;
- (b) Simulation of configurations containing from 1 up to 5 piezoelectric patch pairs;
- (c) Discretization of the patch positioning domain with a step size of up to 0.01 in the x and y directions;
- (d) Determination of the optimal positions and number of patches for both low- and high-frequency modes.

2. Harmonic Analysis

- (a) Implementation of a single shunt-type electrical circuit in a parallel configuration;
- (b) Calculation of RL parameters using Equation 2.13;
- (c) Simulation of the harmonic response considering a $\pm 10\%$ variation around the natural frequency of each analyzed mode;
- (d) Evaluation of control effectiveness for each optimal configuration and comparison between the different simulated scenarios.

In this first approach, the optimization was carried out exclusively during the modal analysis stage, using the DMS program to solve the proposed problem. To this end, the following multi-objective optimization problem (MOO) was formulated:

$$\begin{aligned}
 &\text{Minimize} \quad F(\mathbf{x}) = \left(-k_{e(m)}^2, -N_{\text{patch pairs}} \right) \\
 &\text{Subject to:} \\
 &\quad \mathbf{x} = [x_1, y_1, \dots, x_5, y_5]^T \\
 &\quad x_n \in [0 : \text{increment} : 1, -1] \\
 &\quad y_n \in [0 : \text{increment} : 1, -1] \\
 &\quad \text{for } n = 1, 2, 3, 4, 5 \\
 &\quad \text{for } m = 1, 2, \dots, 10
 \end{aligned} \tag{4.1}$$

where m represents each of the analyzed modes, and (x, y) defines the position of each patch pair on the plate. The value -1 indicates that the respective pair is inactive, allowing the algorithm to decide not only the best positions but also the optimal number of patch pairs to be used.

4.1.2 Modal Analysis

Solving the optimization problem defined in Equation (4.1), Figure 4.1 below shows the highest values of $k_{e(m)}^2$ obtained for each of the first 10 natural modes analyzed, considering the optimal positions of the pairs of patches in each simulated configuration.

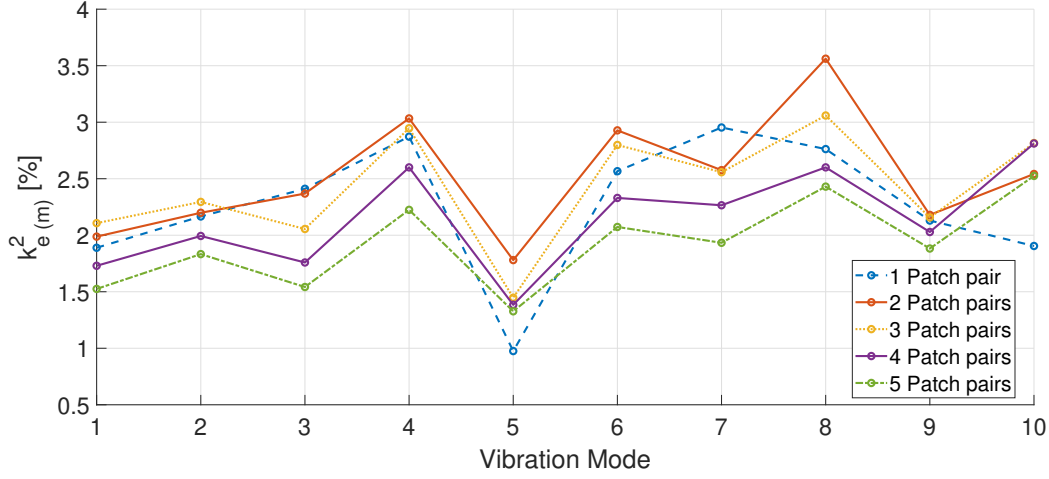


Figure 4.1: Distribution of the values of $k_{e(m)}^2$ related to the first 10 natural modes, with pairs of patches positioned in their respective optimal configurations for each analyzed quantity.

Based on the results presented in Figure 4.1, it is observed that, among the 10 modes analyzed, configurations with 1 to 3 pairs of patches resulted in the highest values of $k_{e(m)}^2$. This indicates that distributions with 4 or more pairs are not effective, since the electromechanical coupling in these configurations tends to decrease, reflected in the reduction of $k_{e(m)}^2$ values. This performance drop may be related to the fact that, by keeping the total mass of the patches constant, dividing it into more units results in increasingly smaller patches, which may cause less individual influence on the coupling with the mechanical system of the structure.

In order to illustrate some of the optimal distributions obtained in the previous step, the following figures show the displacement in the z direction of the structure on the xy plane, considering vibration modes 1 - Figure 4.2 - and 5 - Figure 4.3. For each of these modes, the optimal configurations varying from 1 to 5 pairs of piezoelectric patches are shown.

A relevant point to highlight from the results presented so far refers to the maximum values of $k_{e(m)}^2$ obtained for mode 5. As illustrated in Figure 4.1, it is observed that, even with the optimization of patch positions, the values of $k_{e(m)}^2$ for this mode decrease regardless of the number of pairs of patches applied to the structure.

This behavior seems to be more related to an inherent characteristic of the vibration mode itself than to a limitation of the optimization algorithm. Considering that, when analyzing the optimal patch distributions shown in Figure 4.3, it is seen that the mass distribution of the piezoelectric patches significantly influences the modal pattern of the structure, shifting the nodes and displacement peaks. This sensitivity is not unexpected, given that the mass distribution on the structure directly affects its modal properties.

Consequently, although the algorithm seeks to position the patches in regions of highest modal displacement (peaks/valleys), this dynamic relocation makes it difficult to achieve efficient electromechanical coupling. For this reason, specifically in mode 5, a tendency to reduce the values of $k_{e(m)}^2$ is observed even in the configurations considered optimal.

Therefore, focusing on the configurations that showed the best EMCC results — those with 1, 2, and

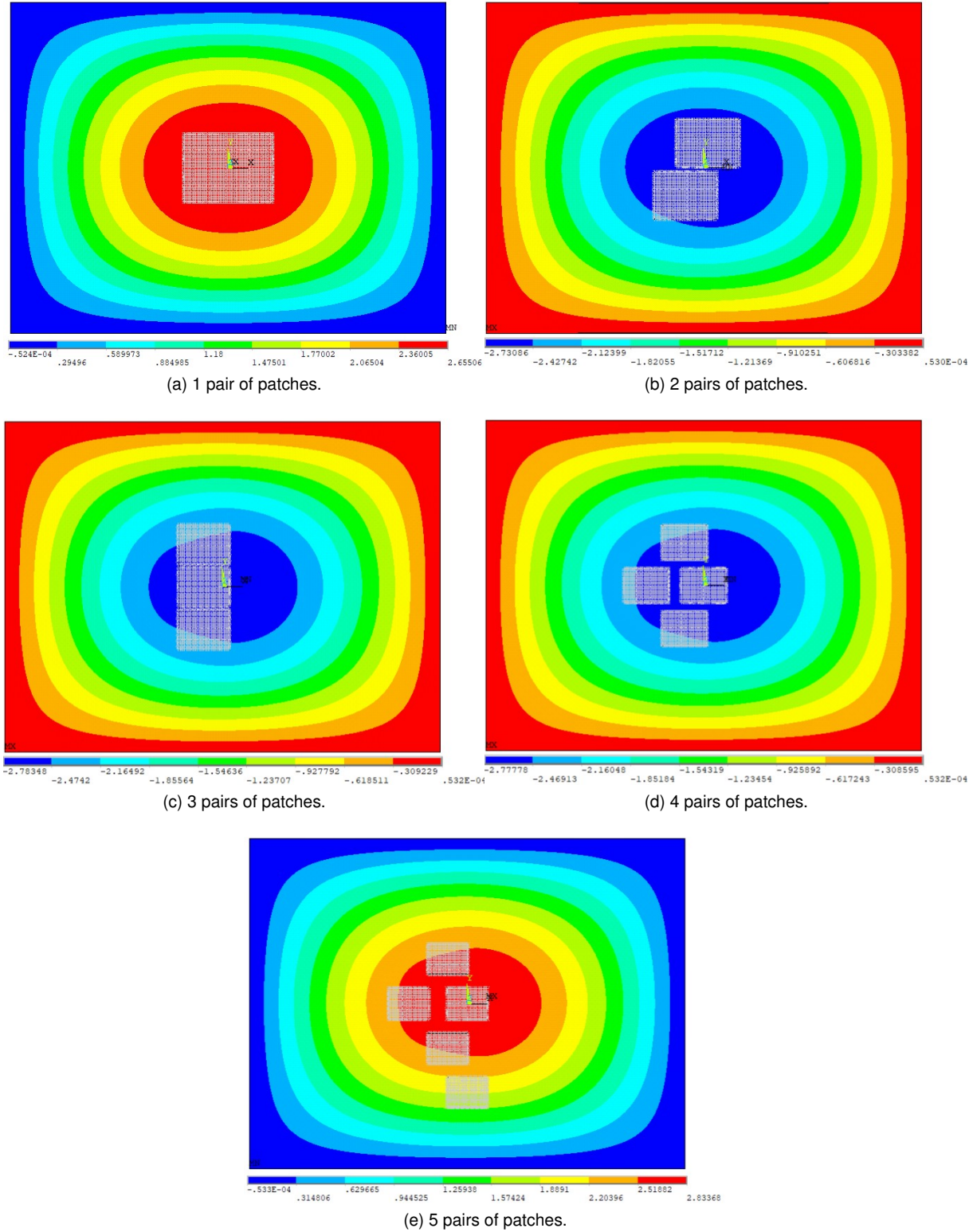
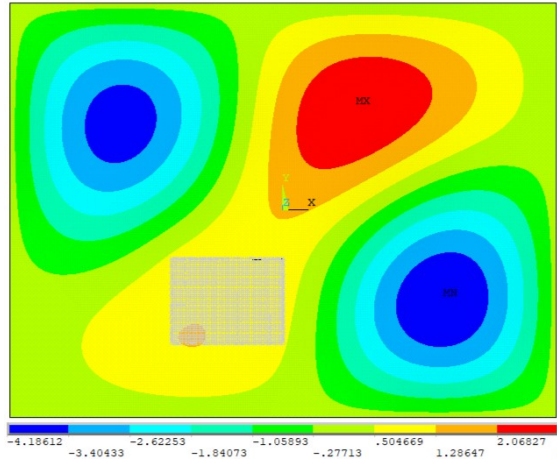
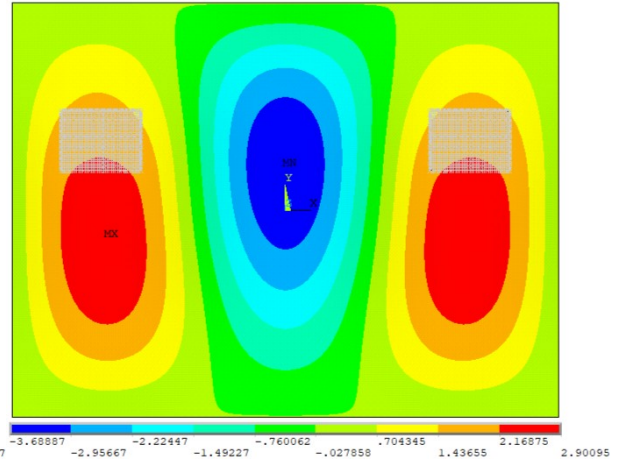


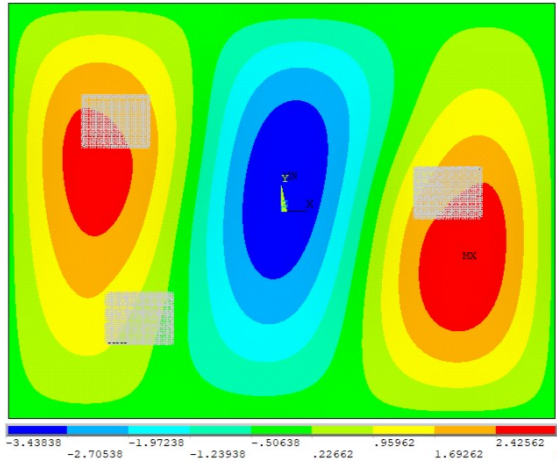
Figure 4.2: Optimal positioning solutions of pairs of piezoelectric patches on the plate structure for the first vibration mode, considering configurations with 1 to 5 pairs.



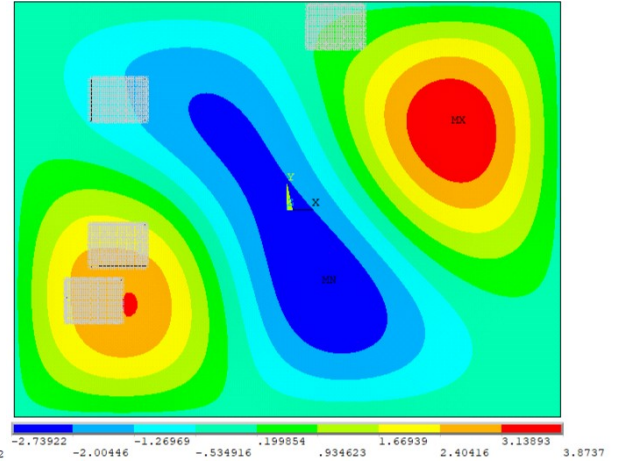
(a) 1 pair of patches.



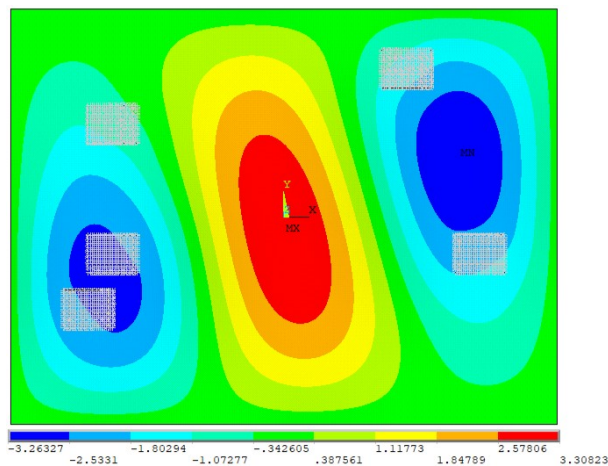
(b) 2 pairs of patches.



(c) 3 pairs of patches.



(d) 4 pairs of patches.



(e) 5 pairs of patches.

Figure 4.3: Optimal positioning solutions of pairs of piezoelectric patches on the plate structure for the fifth vibration mode, considering configurations with 1 to 5 pairs.

3 pairs of piezoelectric patches — Tables 4.1, 4.2, and 4.3 below present a summary of the main data obtained for each of these optimal configurations, respectively, considering the different modes analyzed. These results include the ideal positions of each patch pair, the associated natural frequencies, the modal charges, as well as the values of the RL electrical components calibrated based on equation 2.13, corresponding to each of the optimal distributions found.

Table 4.1: Local minimum calculated by the DMS program for 1 pair of patches.

Modes	Position		Results					
	x_1	y_1	f_{OC}	f_{SC}	k_e^2	Q_{SC}	L_p	R_p
	[%]		[Hz]	[Hz]	[%]	[mCs]	[H]	[k Ω]
1	50	50	36.16	35.82	1.89	22.84	36.56	42.13
2	20	50	80.03	79.18	2.17	52.35	7.88	18.86
3	50	80	112.63	111.29	2.41	80.23	3.76	11.96
4	90	20	152.57	150.43	2.87	117.34	2.08	8.22
5	37	22	156.41	155.65	0.98	70.39	1.98	13.81
6	50	85	227.86	224.99	2.57	168.38	0.91	5.65
7	40	50	244.75	241.22	2.95	193.09	0.79	4.94
8	5	50	264.92	261.34	2.76	199.10	0.70	4.86
9	80	50	286.76	283.76	2.13	188.59	0.60	5.17
10	66	80	335.77	332.69	1.86	203.97	0.45	4.85

Table 4.2: Local minimum calculated by the DMS program for 2 pairs of patches.

Modes	Position				Results				
	x_1	y_1	x_2	y_2	f_{OC}	f_{SC}	k_e^2	Q_{SC}	L_p
	[%]				[Hz]	[Hz]	[%]	[mCs]	[H]
1	51	59	45	40	36.44	36.09	1.99	23.56	35.11
2	80	50	20	50	77.36	76.53	2.20	52.57	7.89
3	50	30	50	10	113.60	112.28	2.37	80.12	3.68
4	85	25	10	75	146.46	144.28	3.03	116.03	2.26
5	90	70	10	70	153.79	152.44	1.78	94.42	2.00
6	50	85	10	15	220.17	217.02	2.93	172.01	0.99
7	40	90	40	10	237.65	234.70	2.54	174.18	0.83
8	65	50	35	50	257.81	253.34	3.56	218.21	2.17
9	80	90	80	10	278.25	275.26	2.18	185.10	0.64
10	90	15	5	85	326.94	322.86	2.54	242.61	0.43

Based on the results presented in tables 4.1, 4.2, and 4.3, two important points can be highlighted. The first refers to the low-frequency modes, such as modes 1, 2, and 3. In these cases, the natural frequencies are considerably spaced. However, as we move to higher frequency modes, like modes 6, 7, and 8, a significant closeness between the natural frequencies is observed, indicating a modal behavior that will be discussed in more depth in the following sections due to its relevance to subsequent results in this thesis.

Moreover, another relevant aspect relates to the values of the calibrated RL electrical parameters. Even with variations in the number of patch pairs, the calibrated RL values remain quite close across most analyzed modes. This behavior is consistent since, with patches positioned in their optimal locations, it is expected that the values of $k_{e(m)}^2$ and the associated modal charges do not show large

Table 4.3: Local minimum calculated by the DMS program for 3 pairs of patches.

Modes	Position						Results					
	x_1	y_1	x_2	y_2	x_3	y_3	f_{OC}	f_{SC}	k_e^2	Q_{SC}	L_p	R_p
			[%]				[Hz]	[Hz]	[%]	[mCs]	[H]	[k Ω]
1	45	65	45	50	45	35	36.77	36.39	2.11	24.43	35.18	39.25
2	80	50	30	55	15	55	78.35	77.47	2.29	54.22	7.77	17.70
3	40	35	40	20	30	80	111.32	110.20	2.05	73.06	3.83	13.11
4	85	25	15	65	10	80	146.66	144.55	2.95	114.41	2.25	8.41
5	85	55	20	20	15	75	152.11	151.02	1.45	78.27	2.36	13.18
6	60	50	40	90	40	10	229.42	226.28	2.80	175.06	0.91	5.48
7	60	50	50	90	50	10	232.71	229.80	2.56	170.20	0.88	5.63
8	90	50	10	55	5	40	254.48	250.69	3.04	201.58	0.75	4.78
9	75	60	75	45	30	90	281.03	278.04	2.16	188.29	0.61	5.12
10	90	15	60	15	5	85	321.10	316.68	2.82	246.42	0.46	3.88

variations, resulting in similar electrical parameters across different configurations.

4.1.3 Harmonic Analysis

With the optimal positions previously defined for each configuration of patch pair quantity, the calibrated RL values were calculated and are presented in Tables 4.1, 4.2, and 4.3. Considering only the configurations with 1 to 3 patch pairs—which yielded the best EMCC results—and using the corresponding RL values, harmonic analysis was performed for the low-frequency modes (modes 1, 2, and 3) as well as for one high-frequency mode (mode 7), considering a variation range of $\pm 10\%$ around the natural frequency corresponding to each mode.

Additionally, the importance of properly selecting the force application points in harmonic analysis is emphasized. As widely recognized, the effective excitation of a specific mode depends directly on the location of the applied force. For this reason, the excitation points were strategically defined for each mode analyzed, as will be detailed in the following sections. The magnitude of the applied force was fixed at 100 N, as specified in Table 3.1.

Low-Frequency Modes

For each of the first three vibration modes, harmonic analyses were carried out with the objective of assessing the effectiveness of the passive shunt control strategy. The simulations were initially performed using the previously calibrated RL values, which were determined based on the optimal patch configurations and the modal parameters of each mode. These RL values aim to maximize the energy transfer between the mechanical and electrical domains, thus enhancing vibration attenuation. Subsequently, the same scenarios were simulated using RL values approaching zero. This setup represents a baseline or reference condition in which the piezoelectric patches are effectively inactive.

By comparing the results from both sets of simulations, it becomes possible to clearly identify the impact of implementing the control system, particularly in terms of reducing the peak amplitudes of vibration. The outcomes of these analyses are presented and discussed in the following items:

• 1st Vibration Mode

For the first vibration mode, a point force was applied at the position $(x; y) = (0,0000; 0,0000)$, which was also used as the observation point for measuring the amplitude in the harmonic responses shown in the following figures. Under this condition, the harmonic responses shown in Figure 4.4 were obtained.

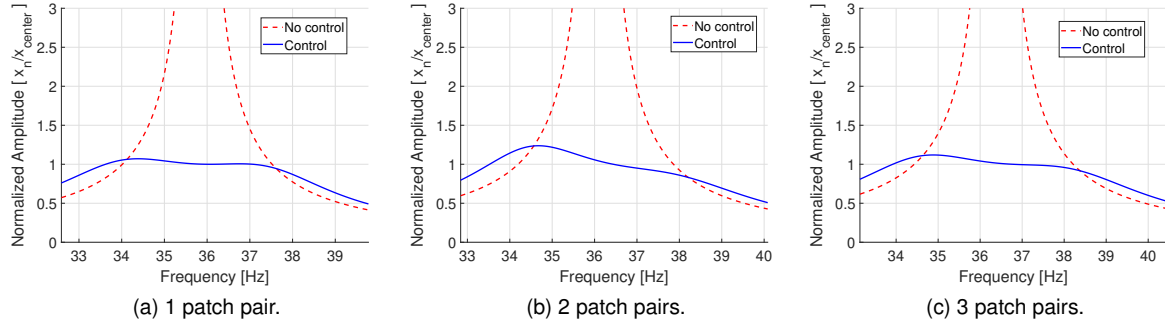


Figure 4.4: Harmonic solutions for the 1st vibration mode using a single shunt circuit with calibrated RL values.

The normalization parameter, defined as x_{center} , corresponds to the displacement amplitude obtained when the excitation frequency matches the natural frequency of the analyzed vibration mode. It can be observed that in all cases, there was a reduction in the vibration peak with the application of the shunt control. To evaluate which configuration showed the highest efficiency in vibration mitigation, Figure 4.5 presents a direct comparison between the obtained results.

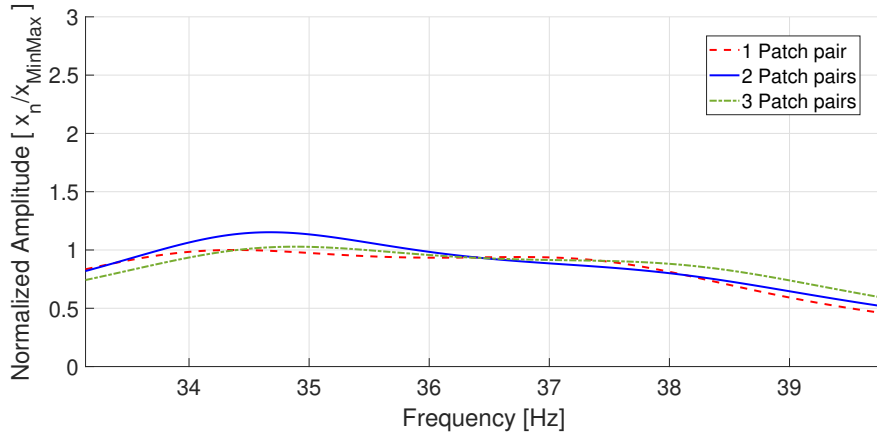


Figure 4.5: Comparative analysis of the harmonic responses in the 1st vibration mode, considering calibrated RL values for configurations with 1, 2, and 3 pairs of piezoelectric patches.

In Figure 4.5, the normalization parameter used is x_{MinMax} , defined as the amplitude corresponding to the peak of the curve that showed the lowest value among all the analyzed configurations. This normalization enables a relative comparison with respect to the best response obtained in terms of vibration reduction. Based on the presented results, it can be observed that all optimal configurations were effective in mitigating the vibration peak, with none standing out significantly over the others.

• 2nd Vibration Mode

For the second vibration mode, a single point force was again applied, now positioned at $(x; y) = (-0.0975; 0.0000)$, which was also used for the response amplitude reading. The harmonic responses corresponding to this mode are presented in Figure 4.6.

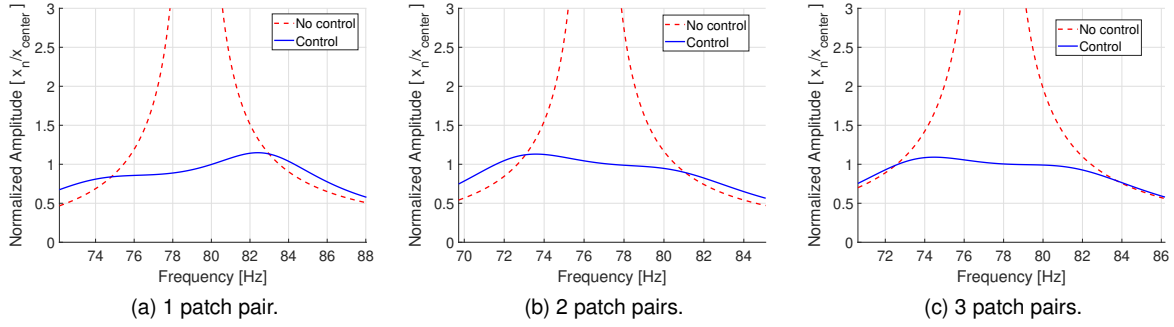


Figure 4.6: Harmonic solutions for the 2nd vibration mode using a single shunt circuit with calibrated RL values.

As observed in the case of the 1st vibration mode, the application of the shunt control also proved effective in the different configurations tested for the 2nd mode. Aiming to compare the performance among these configurations, Figure 4.7 presents the resulting curves of the normalized harmonic responses.

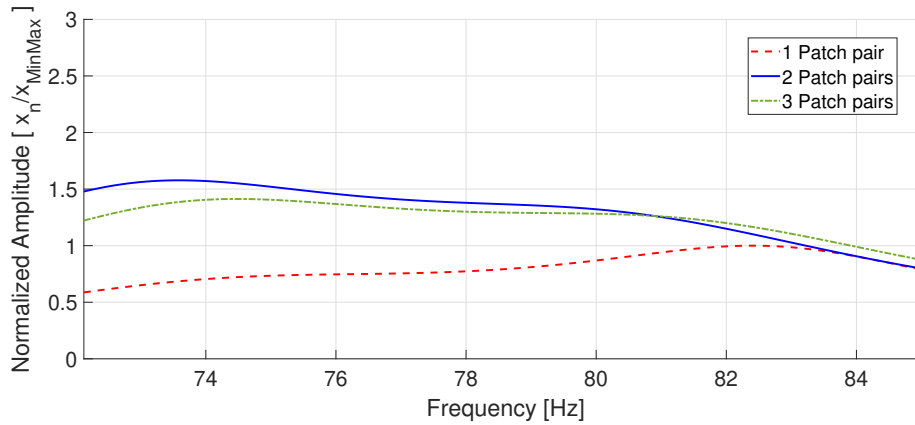


Figure 4.7: Comparative analysis of harmonic responses in the 2nd vibration mode, considering calibrated RL values for configurations with 1, 2, and 3 pairs of piezoelectric patches.

As can be seen in Figure 4.7, the configuration with 1 patch pair showed the most effective reduction in the vibration peak among the tested alternatives, even though it presented a lower EMCC value compared to the others, as indicated in Figure 4.1. This behavior may be related to the reduction in patch size in configurations with a higher number of patches, which directly influences the system's modal charge. However, it is not possible to definitively conclude that one configuration is superior to the others, since the resistance and inductance values calculated during calibration also play a decisive role. Therefore, the use of optimized electrical parameters, rather than merely calibrated ones, could modify these responses and lead to different conclusions.

• 3rd Vibration Mode

Finally, for the third vibration mode, the point force was repositioned to $(x; y) = (0,0000; 0,0735)$, which is also the point used to measure the response amplitude. The harmonic responses obtained for this configuration are shown in Figure 4.8.

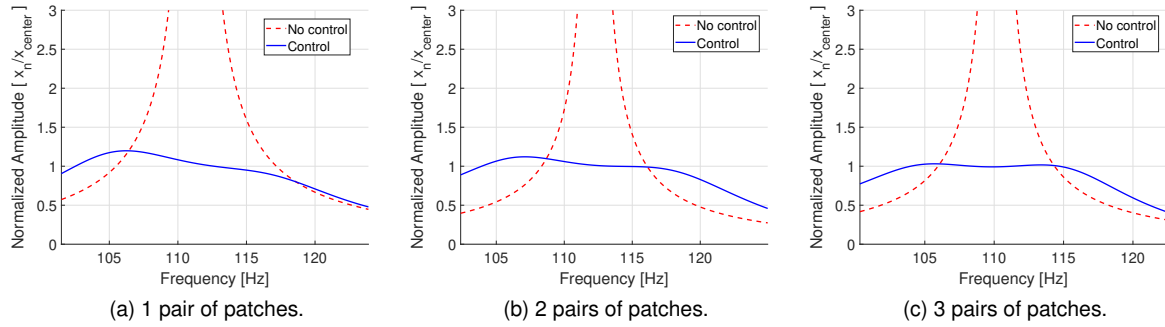


Figure 4.8: Harmonic solutions for the 3rd vibration mode using a single shunt circuit with calibrated RL values.

As can be seen in Figure 4.8, all shunt control configurations effectively reduced the vibration peak. Figure 4.9 presents a comparative analysis of these different configurations.

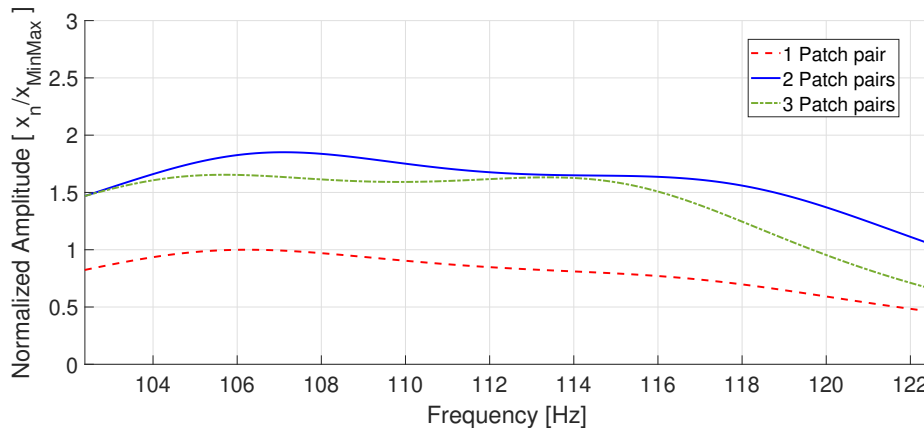


Figure 4.9: Comparative analysis of harmonic responses for the 3rd vibration mode, considering calibrated RL values for configurations with 1, 2, and 3 pairs of piezoelectric patches.

As observed in the second vibration mode, Figure 4.9 shows that the configuration with only 1 pair of piezoelectric patches was also the most effective for the third mode. In this case, the reduction in the vibration peak was significantly greater compared to the other configurations, reaching almost half the amplitude obtained with 2 and 3 pairs.

Based on the results obtained for the low-frequency modes (modes 1, 2, and 3), it can be concluded that when using the RL values calibrated by equation 2.13, the configuration with 1 pair of piezoelectric patches proves to be more effective. This superiority may be associated, among other factors, with the more pronounced extent of the peaks and valleys in the low-frequency modes, which facilitates electromechanical coupling with fewer actuators. Additionally, the greater separation between the natural

frequencies of these modes allows for individual optimization of each mode without significant interference from adjacent modes, thereby enhancing the efficiency of the shunt control even with a reduced number of patches.

High-Frequency Modes

Shifting the focus of the analysis to high-frequency modes, the seventh vibration mode was selected as representative for this study. As was done for the low-frequency modes, harmonic analyses were conducted for the three configurations with 1, 2, and 3 pairs of piezoelectric patches, both with the implementation of passive shunt control and under an uncontrolled condition, using an analysis range of $\pm 10\%$ around the natural frequency of the mode under investigation.

Before presenting the results, it is important to justify the division of the study between low- and high-frequency modes. As discussed in Section 4.1.2 and evidenced in Tables 4.1, 4.2, and 4.3, as frequency increases, the natural modes tend to become closer in terms of frequency, resulting in vibration peaks that are more narrowly spaced.

Given this context, the analysis of high-frequency modes requires attention not only to the target mode (in this case, mode 7), but also to adjacent modes that may be excited within the analysis range. For this reason, unlike previous analyses that used a single point force, two point forces of the same magnitude and orientation were applied in this case, as defined beforehand. The first force was strategically placed to primarily excite the seventh mode and was also used as the measurement point for the displacement amplitude in the z direction. The second force was applied at a position capable of effectively exciting the adjacent modes (modes 6 and 8), which have close natural frequencies and can therefore interfere with the response.

Based on these criteria, the forces were applied at the following positions:

$$(x_{F_1}; y_{F_1}) = (0,0360; 0,0000) \quad (x_{F_2}; y_{F_2}) = (0,0000; 0,1085)$$

The harmonic responses obtained for this configuration are shown in Figure 4.10.

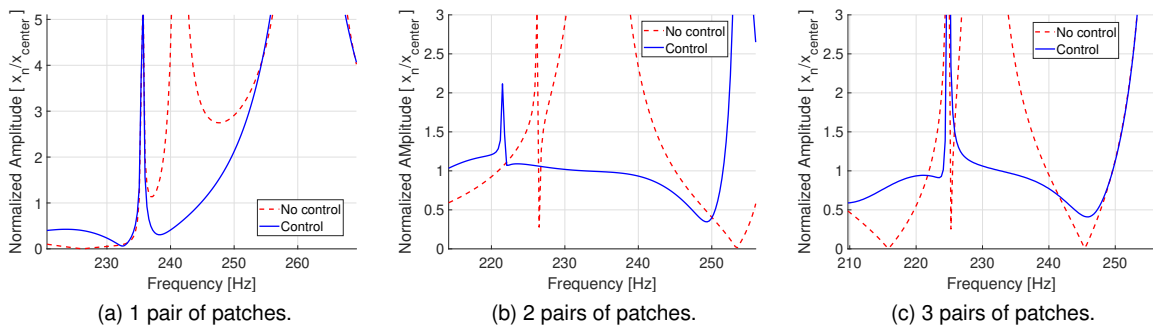


Figure 4.10: Harmonic solutions for the 7th vibration mode using a single shunt circuit with calibrated RL values.

As can be seen in the plots in Figure 4.10—and as already expected—when analyzing the behavior of the 7th vibration mode, the presence of adjacent modes within the considered frequency range is

clearly noticeable. This conclusion can be drawn from the continuous blue curves in the three plots of the aforementioned figure. In each of these curves, three vibration peaks are identified, which coincide with the natural frequencies of modes 6, 7, and 8, as presented in Tables 4.1, 4.2, and 4.3. Therefore, it can be affirmed that these peaks result from the simultaneous excitation of these adjacent modes.

In this context, the results demonstrate that, unlike what was observed for the low-frequency modes, the simple passive shunt control (single shunt control) applied to the smart structure was not effective throughout the entire analyzed frequency range. This behavior can be attributed mainly to three factors:

1. **Unimodal positioning:** the patches were optimized only for mode 7, ensuring good electromechanical coupling (EMCC) for this mode, but potentially resulting in poor coupling in the adjacent modes.
2. **RL values specific to a single mode:** the resistance and inductance values were calibrated exclusively for mode 7, and are not effective for modes 6 and 8.
3. **Modal charges with opposite signs:** in adjacent modes, patches may exhibit charge distributions with opposite signs, causing cancellations that reduce the effectiveness of the control.

Given the points listed above, the solutions proposed in this thesis to address each of the identified problems are as follows:

1. **Unimodal positioning** → **Multi-objective positioning optimization**

Aims to simultaneously consider multiple modes when selecting patch positions, improving overall performance.

2. **RL values specific to a single mode** → **RL parameter optimization**

Involves fine-tuning the electrical parameters to maximize performance over a broader frequency range.

3. **Modal charges with opposite signs** → **Application of multishunt control**

Allows for handling conflicting modal distributions through independent shunt circuits.

4.2 Multimodal Approach

4.2.1 Problem Description

As presented in the previous section, the unimodal approach proved to be ineffective for attenuating vibrations in high-frequency modes. To overcome this limitation, a new strategy was adopted: the multimodal approach, which considers not only the target mode but also the adjacent modes that are excited within the analyzed frequency range.

In this new context, the problem consists of finding the optimal positions for the piezoelectric patch pairs (ranging from 1 to 3 pairs), simultaneously maximizing the electromechanical coupling values $k_{e(6)}^2$, $k_{e(7)}^2$, and $k_{e(8)}^2$, corresponding to modes 6, 7, and 8.

Once the optimal positions are defined, the second step of the solution becomes the optimization of the resistance (R) and inductance (L) values for each circuit. Unlike the previous model, this approach employs a multi-shunt control scheme, in which each patch pair has its own independent shunt circuit, allowing for more effective control in the presence of modal variations.

4.2.2 Solution Flow

As in the unimodal approach, the solution involves two stages: the modal analysis, responsible for optimizing the placement of patch pairs, and the harmonic analysis, aimed at optimizing the RL values. In each of these phases, models capable of overcoming the limitations identified in the unimodal approach were applied.

Modal Analysis

In the multimodal approach, the number of patch pairs is not treated as a decision variable, since, as discussed in Section 4.1.2, it has already been determined that the optimal configurations range from 1 to 3 pairs. The focus, therefore, is on identifying the optimal patch positions for each of these configurations, considering simultaneously the coupling of modes 6, 7, and 8.

This optimization is not trivial, since assigning equal weights to the three objectives leads the solution to no longer be a single optimal point but instead a set of possible optimal solutions known as the Pareto front.

Thus, the multi-objective optimization (MOO) problem solved by the DMS program is formulated as:

$$\begin{aligned}
 &\text{Minimize } F(\mathbf{X}) = \left(-k_{e(6)}^2, -k_{e(7)}^2, -k_{e(8)}^2 \right) \\
 &\text{Subject to:} \\
 &\quad \mathbf{X} = [x_1, y_1, \dots, x_{N_{\text{patch pairs}}}, y_{N_{\text{patch pairs}}}]^T \\
 &\quad x_n, y_n \in [0 : \text{increment} : 1, -1] \\
 &\quad \text{for } N_{\text{patch pairs}} = 1, 2, 3 \quad \text{and } n = 1, \dots, N_{\text{patch pairs}}
 \end{aligned} \tag{4.2}$$

Although the problem can be represented by a single equation as shown above, in practice, it comprises three distinct problems differing only in the number of patch pairs considered. Therefore, the program must find three separate solutions, one for each configuration of patch pair quantity.

Solving each of these problems results in a corresponding Pareto front. To select a point from each front, five different criteria were adopted, each potentially leading to a different solution from the front:

- **Maximum Harmonic Mean**

This approach seeks to balance the values among the objectives, penalizing extreme discrepancies and ensuring that none of the objectives is significantly worse than the others. It is a good strategy when all objectives are of similar importance, as is the case here, as it promotes consistent performance among them and avoids solutions that favor one objective at the expense of the

others. It is mathematically expressed as:

$$H_j = \left(\frac{1}{k_{e,j(6)}^2} + \frac{1}{k_{e,j(7)}^2} + \frac{1}{k_{e,j(8)}^2} \right)^{-1} \text{ for } j = [1, 2, \dots, N_{front}] \quad (4.3)$$

$$\mathbf{X}_{opt} = \mathbf{X}_k \text{ where } H_k = \max(H_1, \dots, H_{N_{front}})$$

where the subscript j refers to the j -th point on the Pareto front, $k_{e,j(m)}^2$ represents the effective EMCC value for mode m at the j -th point, and N_{front} is the total number of points on the front. The optimal point $\mathbf{X}_{opt}$ corresponds to the k -th point with the highest harmonic mean H_k among all calculated means.

- **Minimum Euclidean Distance to the Mean**

This method identifies the point on the Pareto front closest to the average of all points, ensuring that all objectives are, at a minimum, as close as possible to their respective mean values without significantly compromising any of them. This is described mathematically as:

$$\begin{aligned} \mu_6 &= \frac{1}{N_{front}} \sum_{j=1}^{N_{front}} k_{e,j(6)}^2 \\ \mu_7 &= \frac{1}{N_{front}} \sum_{j=1}^{N_{front}} k_{e,j(7)}^2 \\ \mu_8 &= \frac{1}{N_{front}} \sum_{j=1}^{N_{front}} k_{e,j(8)}^2 \end{aligned} \quad (4.4)$$

$$E_j = \sqrt{\left(k_{e,j(6)}^2 - \mu_6\right)^2 + \left(k_{e,j(7)}^2 - \mu_7\right)^2 + \left(k_{e,j(8)}^2 - \mu_8\right)^2}$$

$$\text{for } j = [1, 2, \dots, N_{front}]$$

$$\mathbf{X}_{opt} = \mathbf{X}_k \text{ where } E_k = \min(E_1, \dots, E_{N_{front}})$$

where μ_6 , μ_7 , and μ_8 are the average values of the respective EMCC objectives, and the optimal point $\mathbf{X}_{opt}$ corresponds to the k -th point with the smallest Euclidean distance E_k to the calculated means.

- **Minimum Euclidean Distance to the Median**

Similar to the mean method, this criterion seeks the point closest to the median of the objective values. It reduces the influence of outliers, being more robust in the presence of extreme values. Mathematically:

$$E_j = \sqrt{\left(k_{e,j(6)}^2 - \text{Med}\left(\left\{k_{e,i(6)}^2\right\}_{i=1}^{N_{front}}\right)\right)^2 + \left(k_{e,j(7)}^2 - \text{Med}\left(\left\{k_{e,i(7)}^2\right\}_{i=1}^{N_{front}}\right)\right)^2 + \left(k_{e,j(8)}^2 - \text{Med}\left(\left\{k_{e,i(8)}^2\right\}_{i=1}^{N_{front}}\right)\right)^2}$$

$$\text{for } j = 1, 2, \dots, N_{front}$$

$$\mathbf{X}_{opt} = \mathbf{X}_k \text{ where } E_k = \min(E_1, \dots, E_{N_{front}}) \quad (4.5)$$

where, unlike the previous method, the optimal point $\mathbf{X}_{opt}$ is the configuration with the smallest Euclidean distance E_k relative to the medians of each set.

- **MinMax**

This criterion selects the point on the Pareto front that shows the best value among the worst performances of each solution. In other words, the worst of the three k_e^2 values is considered at each point, and the one whose worst value is better than the others is chosen. It is ideal for ensuring good minimum performance across all objectives.

$$M_j = \max \left(k_{e,j(6)}^2, k_{e,j(7)}^2, k_{e,j(8)}^2 \right) \quad \text{for } j = 1, 2, \dots, N_{\text{front}} \quad (4.6)$$

$$\mathbf{X}_{\text{opt}} = \mathbf{X}_k \quad \text{where} \quad M_k = \min (M_1, \dots, M_N)$$

- **MaxMin**

Finally, this method is the opposite of the previous criterion. Here, the goal is to find the point where the lowest of the three objectives is as high as possible. It ensures that the "worst" performance among the three modes is still the best it can be, promoting robustness and balance.

$$m_j = \min \left(k_{e,j(6)}^2, k_{e,j(7)}^2, k_{e,j(8)}^2 \right) \quad \text{for } j = 1, 2, \dots, N_{\text{front}} \quad (4.7)$$

$$\mathbf{X}_{\text{opt}} = \mathbf{X}_k \quad \text{where} \quad m_k = \max (m_1, \dots, m_N)$$

By applying each of these methods, it is possible to find potential optimal solutions for the positioning problem of the patch pairs. Therefore, the solution method can be organized into the following steps:

1. Consideration of modes 6, 7, and 8 of the structure;
2. Independent simulation considering 1 to 3 pairs of piezoelectric patches;
3. Discretization of the patch placement domain with a step size of up to 0.01 in the x and y directions;
4. Definition of the Pareto front for each quantity of patches;
5. Application of criteria to select the optimal point on the Pareto front.

Harmonic Analysis

With the optimal positions previously determined, the solution flow advances to the harmonic response analysis stage, now focusing on optimizing the RL values with the objective of minimizing the vibration peak of the structure.

Unlike the previous step, which involved multiple objectives, this new phase is characterized as a single-objective optimization problem—the minimization of the peak displacement in the z direction. For this purpose, the GLODS algorithm was used, which is suitable for global, single-objective, derivative-free optimization problems. The formulation of the problem can be expressed as follows:

$$\begin{aligned}
& \text{Minimize} \quad F(\mathbf{x}) = d_z(\mathbf{x}) \\
& \text{Subject to:} \\
& \quad \mathbf{x} = [R_1, L_1, \dots, R_{N_{\text{patch pairs}}}, L_{N_{\text{patch pairs}}}]^T \\
& \quad R_n, L_n \in \mathbb{R}^+ \\
& \quad \text{for } N_{\text{patch pairs}} = 1, 2, 3 \quad \text{and } n = 1, \dots, N_{\text{patch pairs}}
\end{aligned} \tag{4.8}$$

where d_z represents the maximum displacement value in the z direction within the analyzed frequency range, and $\mathbf{x} = [R_1, L_1, \dots, R_n, L_n]$ corresponds to the resistor and inductor values of the n pairs of piezoelectric patches used.

It is important to note that, although only one general equation is defined, the optimization is carried out independently for each positioning configuration and number of patches defined. Therefore, for each placement solution obtained in the modal analysis stage, a new optimization of the electrical parameters is performed based on the previously presented equation.

Thus, the solution flow for this stage can be organized into the following phases:

1. Fix the optimal positions of the piezoelectric patches for each of the analyzed quantities;
2. Estimate the initial resistance and inductance values of each independent circuit using the calibration equations;
3. Define the admissible intervals for the optimization variables (R and L) in each considered configuration;
4. Perform the optimization of the resistance and inductance values based on the previously fixed positions, aiming to minimize the vibratory response;
5. Compare the harmonic responses obtained from the different optimal solutions and establish a criterion for selecting the most effective configuration.

4.2.3 Modal Analysis

By solving the optimization problem defined in Equation (4.2), Figures 4.11, 4.12 and 4.13 respectively present the Pareto fronts obtained for the configurations with 1, 2, and 3 pairs of piezoelectric patches. In each front, the points considered optimal according to the different adopted selection criteria are highlighted. For a more detailed analysis of these points, two-dimensional projections of each combination among the $k_{e(m)}^2$ values are also presented, allowing for a clearer visualization of the distribution of the solutions and an assessment of the effectiveness of each metric used for selecting the optimal point.

As observed in the Pareto fronts, as the number of piezoelectric patch pairs increases, so does the number of solutions lying on the front. This behavior is associated with the high complexity and generality of the developed model, in which each pair of piezoelectric patches is treated as an independent element. As a result, the search space expands significantly, leading to a greater diversity of potentially optimal solutions.

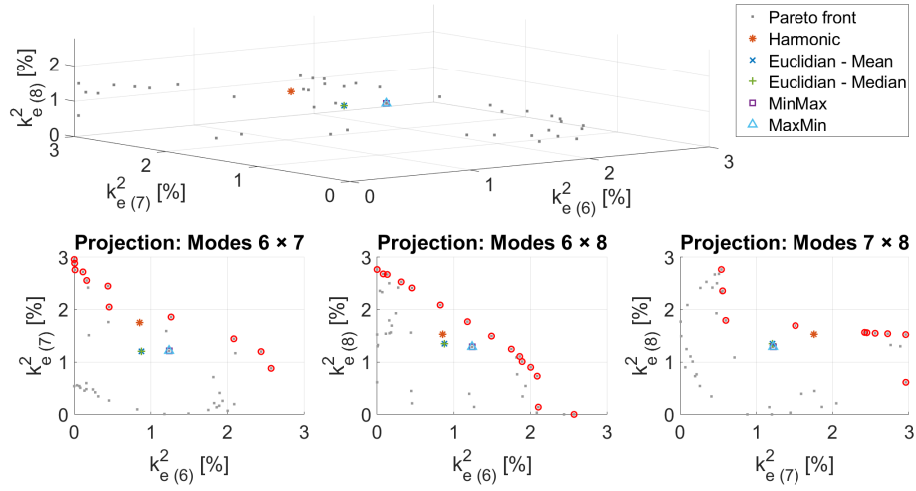


Figure 4.11: Pareto front for the configurations considering 1 pair of piezoelectric patches.

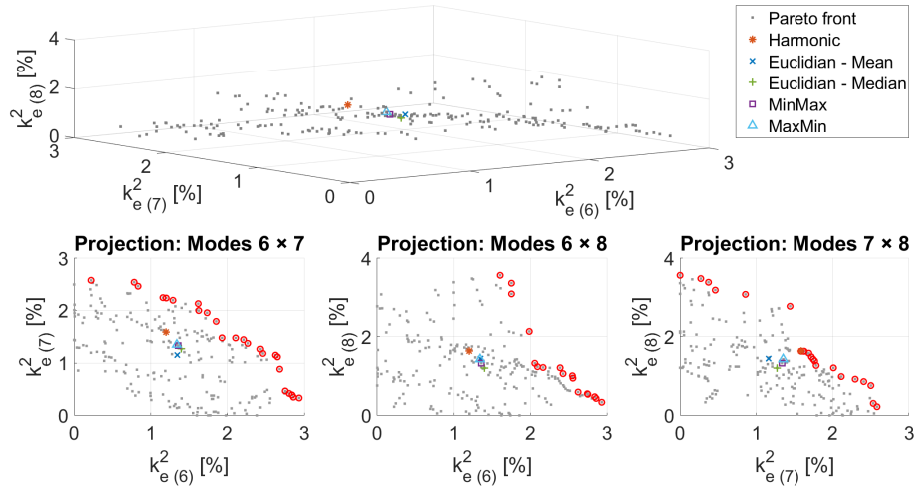


Figure 4.12: Pareto front for the configurations considering 2 pairs of piezoelectric patches.

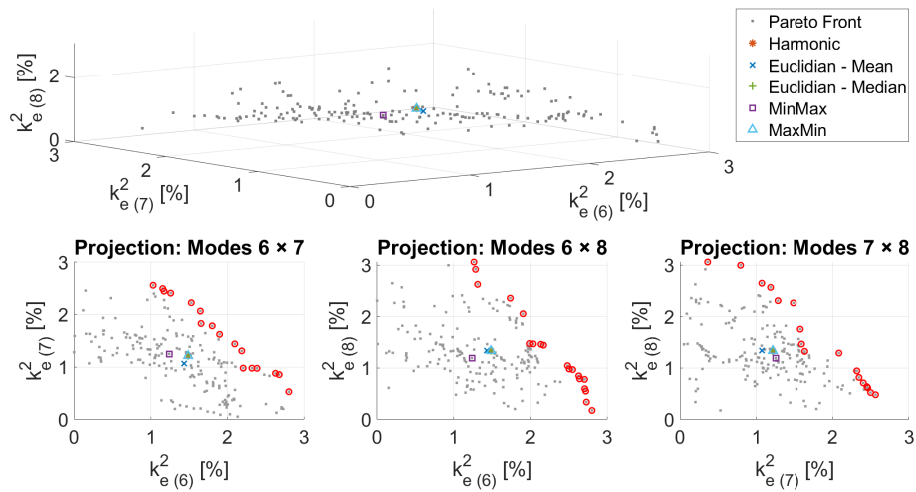


Figure 4.13: Pareto front for the configurations considering 3 pairs of piezoelectric patches.

Furthermore, as evidenced by the 2D projections, the points selected as optimal tend to be located in central regions and near the boundaries in each of the $k_{e(m)}^2$ combinations. This indicates that the adopted selection criteria promote balanced solutions, without overly favoring one objective to the detriment of the others, thus prioritizing consistent performance across the different vibration modes.

The following Tables 4.4, 4.5, and 4.6 detail the characteristics of the possible optimal points selected by the adopted criteria.

Table 4.4: Local minimum calculated using the optimal selection criteria for points on the Pareto front with 1 pair of piezoelectric patches.

Criterion	Position		Results					
	x_1	y_1	$k_{e(6)}^2$	$k_{e(7)}^2$	$k_{e(8)}^2$	$q_{SC(6)}$	$q_{SC(7)}$	$q_{SC(8)}$
	[%]		[%]	[%]	[%]	[mCs]	[mCs]	[mCs]
Harmonic Mean	33	46	0.85	1.75	1.53	-158.20	-95.92	+145.24
Euclidean Dist. (Mean)	31	57	0.89	1.20	1.34	-127.41	-105.33	+137.20
Euclidean Dist. (Median)	31	57	0.89	1.20	1.34	-127.41	-105.33	+137.20
MinMax	31	46	1.24	1.21	1.29	-158.60	-84.74	+133.92
MaxMin	31	46	1.24	1.21	1.29	-158.60	-84.74	+133.92

Table 4.5: Local minimum calculated using the optimal selection criteria for points on the Pareto front with 2 pairs of piezoelectric patches.

Criterion	Position				Results					
	x_1	y_1	x_2	y_2	$k_{e(6)}^2$	$k_{e(7)}^2$	$k_{e(8)}^2$	$q_{SC(6)}$	$q_{SC(7)}$	$q_{SC(8)}$
	[%]				[%]	[%]	[%]	[mCs]	[mCs]	[mCs]
Harmonic Mean	90	85	35	50	1.19	1.58	1.63	-81.44 +17.91	+16.12 +99.43	+73.01 +75.49
Euclidean Dist. (Mean)	90	15	30	55	1.35	1.16	1.44	-88.29 +10.83	-1.61 -85.84	-67.71 -71.76
Euclidean Dist. (Median)	38	54	10	73	1.40	1.27	1.21	+41.56 -81.33	+86.55 -3.92	+63.47 -62.89
MinMax	38	54	8	73	1.36	1.33	1.33	+38.97 -81.87	+88.39 -7.62	-64.52 +68.31
MaxMin	90	15	30	50	1.34	1.35	1.43	-86.31 +19.25	+16.04 +92.21	-69.10 -69.02

As evidenced by the results in the tables, increasing the number of piezoelectric patch pairs generally leads to higher values of $k_{e(m)}^2$ when the three vibration modes are considered together. Moreover, it is observed that as the vibration mode changes, the signs of the modal charges vary, which may compromise the effectiveness of a single shunt control due to potential cancellation between contributions of opposite signs.

Therefore, the next section aims to apply multi-shunt control to at least one of the optimal distributions obtained for each analyzed patch quantity configuration. The selection of these distributions considered

Table 4.6: Local minimum calculated using the optimal selection criteria for points on the Pareto front with 3 pairs of piezoelectric patches.

Criterion	Position						Results					
	x_1	y_1	x_2	y_2	x_3	y_3	$k_{e(6)}^2$	$k_{e(7)}^2$	$k_{e(8)}^2$	$q_{SC(6)}$	$q_{SC(7)}$	$q_{SC(8)}$
			[%]				[%]	[%]	[%]	[mCs]	[mCs]	[mCs]
Harmonic Mean	64	52	53	8	6	19	1.49	1.21	1.34	-29.05 +55.50 -41.12	+55.83 -29.66 -26.90	-56.78 -21.93 -50.35
Euclidean Dist. (Mean)	90	60	60	50	50	10	1.43	1.07	1.33	+28.43 -27.58 +62.46	+15.15 +56.32 -26.94	-64.47 +43.50 +12.11
Euclidean Dist. (Median)	64	52	53	8	6	19	1.49	1.21	1.34	-29.05 +55.50 -41.12	+55.83 -29.66 -26.90	-56.78 -21.93 -50.35
MinMax	40	85	35	50	35	10	1.24	1.25	1.19	+60.11 -37.33 +27.28	+14.00 -40.41 +47.22	+30.11 +61.06 +30.62
MaxMin	64	52	53	8	6	19	1.49	1.21	1.34	-29.05 +55.50 -41.12	+55.83 -29.66 -26.90	-56.78 -21.93 -50.35

not only the uniformity of the $k_{e(m)}^2$ values across the modes but also the achievement of high values in order to maximize the electromechanical coupling.

The selected distributions for analysis are defined as follows:

- **1 patch pair** $\implies$ Optimal point selected by the **MinMax** criterion.
- **2 patch pairs** $\implies$ Optimal point selected by the **MinMax** criterion.
- **3 patch pairs** $\implies$ Optimal point selected by the **Harmonic Mean** criterion.

4.2.4 Harmonic Analysis

Unlike the single shunt approach, in which the resistance and inductance values were obtained directly from calibration equations, in the case of multi-shunt control this approach does not represent the best final solution, but it can be useful as an initial step to estimate reasonable values for these electrical components.

Tables 4.7, 4.8, and 4.9 present the resistance and inductance values calculated for each of the analyzed distributions, using the calibration equations defined in Equation 2.13. Since the multi-shunt criterion is being applied, independent resistance and inductance values were assigned to each patch pair in the distribution, based on its respective individual modal charge — as presented in Tables 4.4, 4.5, and 4.6.

It is important to note that, as shown in the previous tables, the modal charge varies among the excited modes, which implies different resistance and inductance values for each excited mode within

the analyzed frequency range.

Table 4.7: Calibrated resistance and inductance values considering 1 pair of patches in the distribution defined by the MinMax criterion.

Pair reference	$R_{(6)}$ [k Ω]	$R_{(7)}$ [k Ω]	$R_{(8)}$ [k Ω]	$L_{(6)}$ [H]	$L_{(7)}$ [H]	$L_{(8)}$ [H]
1	4.70	16.47	7.45	0.49	1.69	0.72

Table 4.8: Calibrated resistance and inductance values considering 2 pairs of patches in the distribution defined by the MinMax criterion.

Pair reference	$R_{(6)}$ [k Ω]	$R_{(7)}$ [k Ω]	$R_{(8)}$ [k Ω]	$L_{(6)}$ [H]	$L_{(7)}$ [H]	$L_{(8)}$ [H]
1	77.99	15.54	31.96	8.97	1.71	3.19
2	17.67	2085.40	28.52	2.03	229.40	2.85

Table 4.9: Calibrated resistance and inductance values considering 3 pairs of patches in the distribution defined by the Harmonic Mean criterion.

Pair reference	$R_{(6)}$ [k Ω]	$R_{(7)}$ [k Ω]	$R_{(8)}$ [k Ω]	$L_{(6)}$ [H]	$L_{(7)}$ [H]	$L_{(8)}$ [H]
1	144.59	36.88	40.65	17.65	3.89	4.15
2	39.62	130.61	272.50	4.84	13.78	27.83
3	72.15	158.77	51.68	8.81	16.75	5.28

As presented in the previous tables, the resistance (R) and inductance (L) values were indicated with a subscript referring to the vibration mode considered. This subscript therefore indicates that the values were calculated based on the modal charge corresponding to that mode, as defined by the calibration equations.

To estimate the initial resistance and inductance values to be used as a starting point in the optimization model, the following criterion was adopted: for each patch pair configuration, the pair that exhibited the highest modal charge within the analyzed vibration mode was selected. Thus, the R and L values were calculated based on this specific charge. In other words, it was considered that each pair contributes mainly to a dominant mode, being preferentially assigned to the mode in which it exerts the greatest influence.

Based on this criterion, the following initial estimates for resistance and inductance values were defined according to the number of patch pairs:

- **1 Patch Pair**

In the configuration with only one patch pair, the application of multi-shunt control is not feasible, since there is only one circuit – characterizing a single-shunt system. In this case, the single pair must simultaneously act on the three vibration modes. Therefore, and based on Table 4.7, the initial resistance and inductance values were defined as the average of the values obtained for the three modes.

- **2 Patch Pairs**

In the configuration with two patch pairs, multi-shunt control becomes feasible, although it is still not possible for each pair to act exclusively on a single mode, since the number of pairs is smaller than the number of excited modes. In this scenario, based on the modal charges of the MinMax configuration presented in Table 4.5, it is observed that pair 1 has greater influence on mode 7, while pair 2 acts more significantly on modes 6 and 8. Thus, based on Table 4.8, it was defined that the electrical circuit values of pair 1 would correspond to mode 7, and those of pair 2 would be the average between the values of modes 6 and 8.

- **3 Patch Pairs**

With three patch pairs, it is possible to directly assign one pair to each vibration mode. As indicated by the modal charges of the Harmonic Mean configuration presented in Table 4.6, pair 1 proved more effective in mode 7, pair 2 in mode 6, and pair 3 in mode 8. Thus, each circuit was calibrated based on the modal charge corresponding to its assigned mode, according to the data in Table 4.9, ensuring a more specific and balanced distribution of control among the modes.

Table 4.10 below presents and summarizes the resistance and inductance values defined as the starting point for the optimization model, according to the criteria established above.

Table 4.10: Calibrated resistance and inductance values for 1, 2, and 3 pairs of piezoelectric patches considering the multimodal approach.

Number of Pairs	$R_{(1)}$ [k Ω]	$L_{(1)}$ [H]	$R_{(2)}$ [k Ω]	$L_{(2)}$ [H]	$R_{(3)}$ [k Ω]	$L_{(3)}$ [H]
1	9.54	0.97	-	-	-	-
2	15.52	1.71	23.09	2.44	-	-
3	36.88	3.89	39.62	4.84	51.68	5.28

Based on the initial values defined in Table 4.10, and considering the limits observed in Tables 4.7, 4.8 and 4.9, it was possible to establish the search ranges for the resistance (R) and inductance (L) variables to be used in the optimization process.

These ranges were defined in such a way as to encompass most of the values identified for each configuration, thus ensuring an adequate exploration domain for the algorithm. Therefore, for each number of patch pairs, the following boundary values were defined, considering a discretization of the domain of 0.1 k Ω for R and 0.1 H for L :

- **1 Pair of Patches** $\implies R \in [4, 20]$ k Ω , $L \in [0.2, 2.0]$ H
- **2 Pairs of Patches** $\implies R \in [10, 40]$ k Ω , $L \in [1.5, 3.5]$ H
- **3 Pairs of Patches** $\implies R \in [30, 60]$ k Ω , $L \in [3.0, 6.0]$ H

By solving the optimization problem defined in Equation (4.8), using the previously established search ranges, the optimal results presented in Table 4.11 were obtained for each patch-pair configuration.

The following comparisons were then simulated with the aim of performing a more detailed analysis of the control performance:

Table 4.11: Optimized resistance and inductance values obtained by the GLODS program for 1, 2, and 3 pairs of piezoelectric patches considering the multimodal approach.

Number of Pairs	$R_{(1)}$ [k Ω]	$L_{(1)}$ [H]	$R_{(2)}$ [k Ω]	$L_{(2)}$ [H]	$R_{(3)}$ [k Ω]	$L_{(3)}$ [H]
1	5.6	0.7	-	-	-	-
2	24.8	1.6	31.4	1.8	-	-
3	35.2	3.0	34.6	4.4	55.4	4.2

Different Circuit Types

To evaluate the efficiency of the optimized multi-shunt control in comparison with the single-shunt approach under the multimodal framework, the resistance and inductance values of the single-shunt circuit were calculated using the calibration equations, considering the same optimal positions previously defined for each patch-pair configuration in the multimodal approach.

These values were obtained based on the total modal charge of the structure, seeking the connection configuration that presented the least interference among the modes—that is, the one with the smallest cancellation of modal charges. For each excited mode, the corresponding values of R and L were calculated, and the final parameters of the single-shunt circuit were defined as the average of the calibrated values obtained for the three analyzed modes. Table 4.12 below summarizes all the defined parameters.

Table 4.12: Calibrated resistance and inductance values for 2 and 3 pairs of piezoelectric patches considering the multimodal approach and the single-shunt circuit.

Number of Pairs	Modes	f_{OC} [Hz]	f_{SC} [Hz]	k_e^2 [%]	Q_{SC} [mCs]	L_p [H]	R_p [k Ω]	L_{avg} [H]	R_{avg} [k Ω]
2	6	230.00	228.45	1.36	120.82	0.93	8.12	1.04	9.60
	7	237.84	236.27	1.33	96.05	1.45	13.15		
	8	261.55	259.83	1.33	132.87	0.75	7.54		
3	6	226.68	225.01	1.49	125.73	0.94	7.72	0.90	8.23
	7	236.34	234.92	1.21	112.45	0.96	9.10		
	8	256.67	254.97	1.34	129.17	0.80	7.86		

Figures 4.14 and 4.15 below present the simulation results of the system's harmonic response for the configurations with 2 and 3 pairs, respectively. In each case, three distinct conditions are compared: uncontrolled system, system with calibrated single-shunt control, and system with optimized multi-shunt control.

As evidenced by the figures, in the multimodal approach, the application of the optimized multi-shunt circuit proves to be significantly more effective than the use of the calibrated single-shunt circuit. With the multi-shunt, a more efficient attenuation of the vibration peaks corresponding to modes 6, 7, and 8 is observed, reflecting a more targeted and effective control.

Different Numbers of Pairs

Based on the best results obtained for each configuration of patch pairs, as presented in the previous analysis, this section also includes the case with 1 patch pair using the optimized single-shunt circuit.

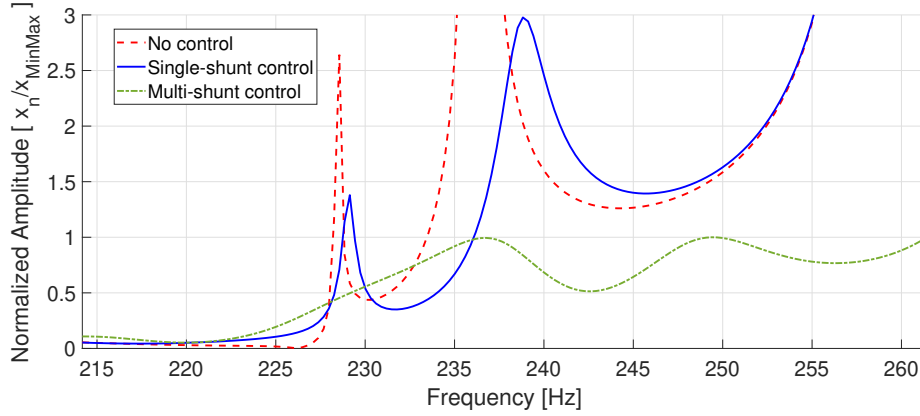


Figure 4.14: Comparison of the system harmonic responses under the following conditions: without control, with calibrated single-shunt control, and with optimized multi-shunt control. Analysis performed within the multimodal approach for the configuration with 2 pairs of piezoelectric patches.

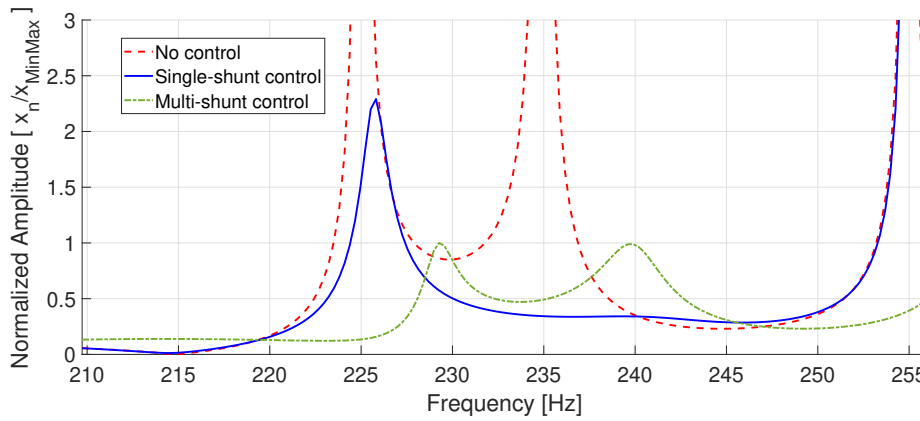


Figure 4.15: Comparison of the system harmonic responses under the following conditions: without control, with calibrated single-shunt control, and with optimized multi-shunt control. Analysis performed within the multimodal approach for the configuration with 3 pairs of piezoelectric patches.

The objective now is to compare, within the multimodal optimization approach, which configuration—in terms of the number of patch pairs—proved to be the most effective in attenuating the vibration peaks.

For this purpose, Figure 4.16 presents the harmonic response corresponding to the following solutions: optimized single-shunt with 1 patch pair, optimized multi-shunt with 2 patch pairs, and optimized multi-shunt with 3 patch pairs.

As can be seen in Figure 4.16, increasing the number of patch pairs results in more effective attenuation of the vibration peaks associated with the modes in the analyzed frequency range. This superior performance is directly related to the use of the multi-shunt circuit, which allows individual tuning of each patch pair to act on different regions of the frequency spectrum.

However, when comparing the cases with 2 and 3 pairs of piezoelectric patches, it is observed that the configuration with 2 pairs outperformed that with 3 pairs—an unexpected result, since it was anticipated that 3 pairs would enable more targeted attenuation of each of the three main vibration modes. This discrepancy may be attributed to effects related to the mass distribution in the structure, which appear to negatively impact the control efficiency in the configuration with 3 patch pairs. This

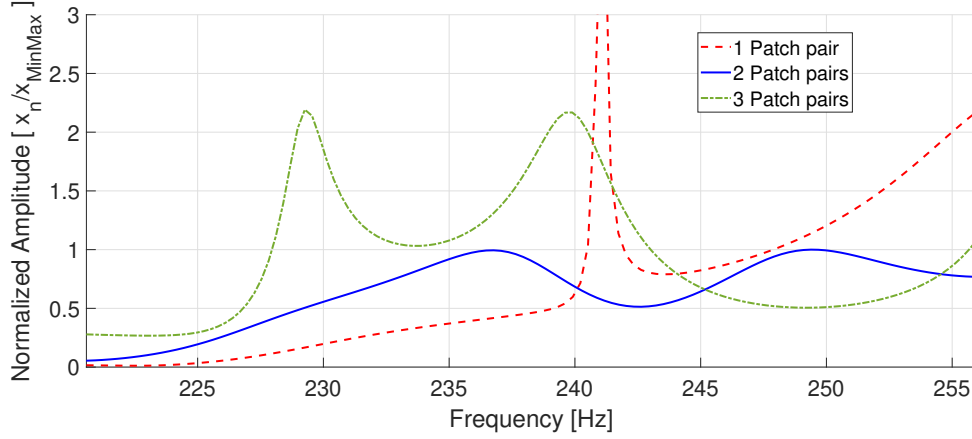


Figure 4.16: Comparison of the system harmonic responses considering the following electrical and geometric configurations: optimized single-shunt with 1 patch pair, optimized multi-shunt with 2 patch pairs, and optimized multi-shunt with 3 patch pairs.

issue will be discussed in more detail in Section 4.4, which presents different comparisons considering the effects of patch distribution on the structure.

4.3 Comparison Between the Approaches

Within the scope of this thesis, two distinct strategies for the application of shunt control in smart structures were studied. The first corresponds to the unimodal approach, in which only the EMCC of the main mode of interest is considered. Based on this premise, the optimal position for placing the piezoelectric patches was determined, and, using the calibration equations presented in Equation 2.13, the resistance and inductance values for the implementation of a single-shunt circuit were calculated.

The second strategy corresponds to the multimodal approach, which takes into account all vibration modes contained within the frequency range of interest. In this case, the placement of the patches was defined so as to provide a satisfactory response for all excited modes. With the positions fixed, the resistance and inductance values were obtained through an optimization process using the GLODS program. Under these conditions, it was possible to implement multi-shunt control for the configurations with 2 and 3 pairs of piezoelectric patches, allowing independent adjustment of the electrical parameters of each circuit.

Therefore, this section aims to compare the performance of the two implemented shunt control approaches. The first, already well established in the literature, is based on the unimodal strategy. The second, developed in the context of this thesis, adopts a multimodal approach focused on the simultaneous attenuation of multiple vibration modes within the same frequency range — in this case, modes 6, 7, and 8. The comparisons for each number of patch pairs are presented in the following subsections.

• 1 Patch Pair

For the configuration with only 1 patch pair, the resistance and inductance values used were those presented in Tables 4.1 (unimodal approach), focusing on mode 7, and 4.11 (multimodal

approach). Figure 4.17 shows the comparison between the simulated harmonic responses, aiming to evaluate which technique was more effective in attenuating the vibration peaks for this configuration.

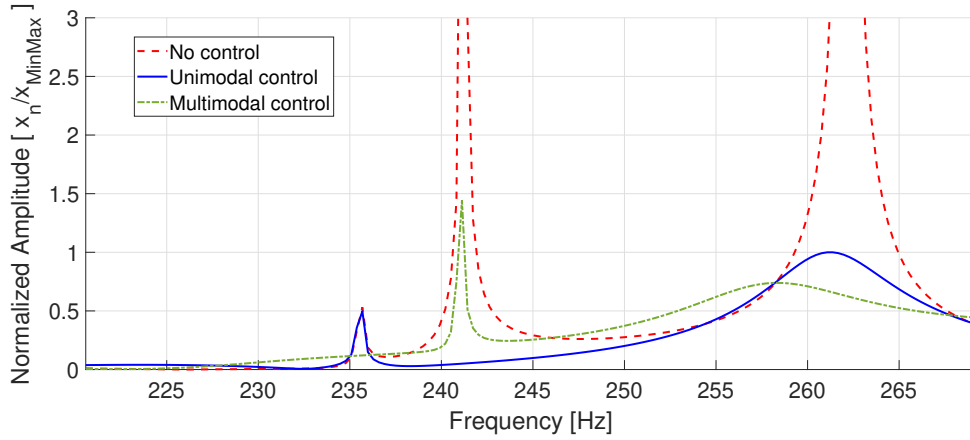


Figure 4.17: Comparison of the harmonic responses of the system with 1 pair of piezoelectric patches under the following conditions: without control, with calibrated single-shunt control (unimodal approach), and with optimized single-shunt control (multimodal approach).

As can be observed in Figure 4.17, the control implemented with the unimodal approach proved more effective in attenuating the vibration peaks compared to the multimodal approach. This is due to the fact that the unimodal approach is specifically targeted at controlling the most critical mode within the analyzed frequency range — in this case, mode 7 — whereas the multimodal approach seeks a more distributed solution among all present modes.

This characteristic means that, although it provides broader action, the multimodal control shows inferior performance in reducing the main peak. Furthermore, although the first peak (around 236 Hz) is not efficiently attenuated by the unimodal approach, its relatively low amplitude makes it less relevant to the overall system performance. Thus, considering the reduction of maximum amplitude as the comparison criterion, the unimodal approach proved more effective in this specific scenario.

• 2 Patch Pairs

Figure 4.18 shows the simulated harmonic responses for the configuration with 2 patch pairs, considering the best results obtained for both approaches. The electrical parameters for the unimodal approach were taken from Table 4.2, focusing on mode 7. Those for the multimodal approach were obtained via optimization, as presented in Table 4.11.

Unlike the case with only 1 patch pair, in this configuration the control using the multimodal approach demonstrated superior performance. This is mainly due to the ability to distribute the patches and tune them to act specifically in different regions of the frequency range, a characteristic that enhances the efficiency of multi-shunt control. In contrast, the unimodal approach, relying on a single circuit tuned for a specific mode, shows more limited performance when multiple relevant modes are present. Therefore, for the configuration with 2 patch pairs, the multimodal approach proved significantly more effective in attenuating the vibration peaks.

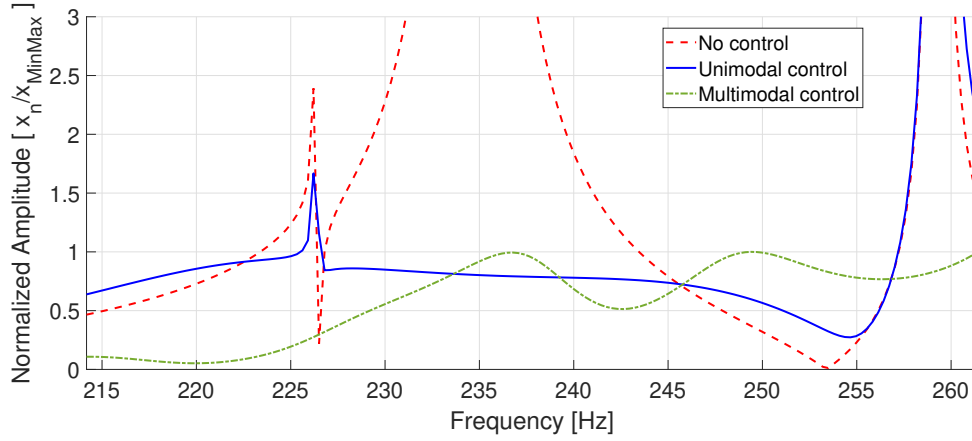


Figure 4.18: Comparison of the harmonic responses of the system with 2 piezoelectric patch pairs under the following conditions: without control, with calibrated single-shunt control (unimodal approach), and with optimized multi-shunt control (multimodal approach).

• 3 Patch Pairs

Finally, Figure 4.19 presents the comparative analysis for the configuration with 3 piezoelectric patch pairs. As in previous cases, the values from Table 4.3 were used for the unimodal approach (focusing on mode 7), and the optimized values from Table 4.11 were used for the multimodal approach.

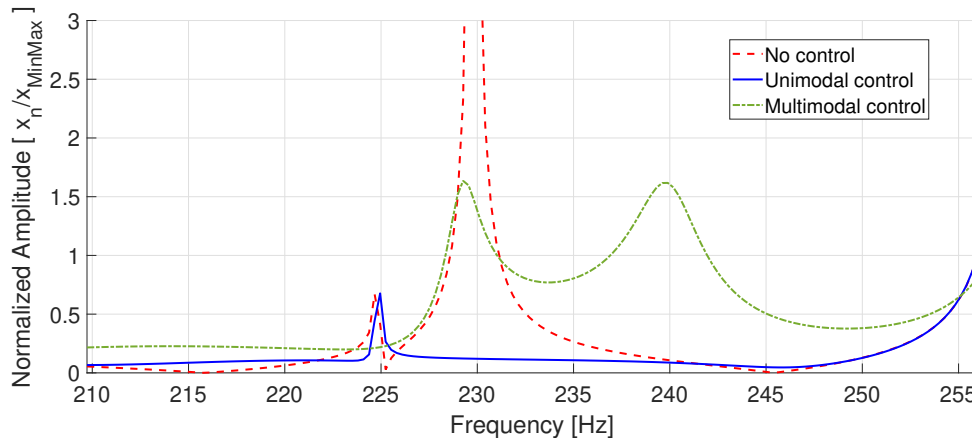


Figure 4.19: Comparison of the harmonic responses of the system with 3 piezoelectric patch pairs under the following conditions: without control, with calibrated single-shunt control (unimodal approach), and with optimized multi-shunt control (multimodal approach).

As illustrated in Figure 4.19, the unimodal approach outperformed the multimodal approach for the configuration with 3 patch pairs, which contradicts the initial expectation. Theoretically, the multimodal control should be more effective in this scenario, since the number of patch pairs matches the number of dominant modes in the analyzed frequency range, allowing each patch to be tuned specifically for a distinct mode. However, the results do not confirm this hypothesis.

This unexpected performance can be attributed to the influence of mass distribution in the structure, which directly affects the system's harmonic response—altering both the amplitude and the

frequency location of the vibration peaks. Although the controls were designed to act over the same frequency range, differences in patch positioning between the approaches may have modified the system's dynamic behavior. This is evidenced by the "No Control" curve, which was obtained from the optimal positioning of the unimodal approach. It can be observed that the resulting unimodal control curve closely follows this reference, whereas the multimodal curve deviates, reflecting an altered vibrational pattern—possibly caused by the redistribution of mass.

Therefore, when using maximum vibration amplitude reduction as the criterion, the unimodal approach demonstrated greater effectiveness in this case. Nevertheless, Section 4.4 will further explore this analysis, examining the impact of mass distribution and discussing the actual effectiveness of each approach in mitigating the total vibrational energy of the system for the configuration with 3 piezoelectric patch pairs.

Thus, after identifying the most effective control approach for each number of patch pairs, it becomes possible to perform a direct comparison between these configurations and evaluate which one provides the lowest maximum amplitude associated with modes 6, 7, and 8 of the structure analyzed in this thesis. Figure 4.20 presents this comparison under the following optimal conditions: calibrated single-shunt control using the unimodal approach with 1 patch pair; optimized multi-shunt control using the multimodal approach with 2 patch pairs; and single-shunt control, also using the unimodal approach, with 3 patch pairs.

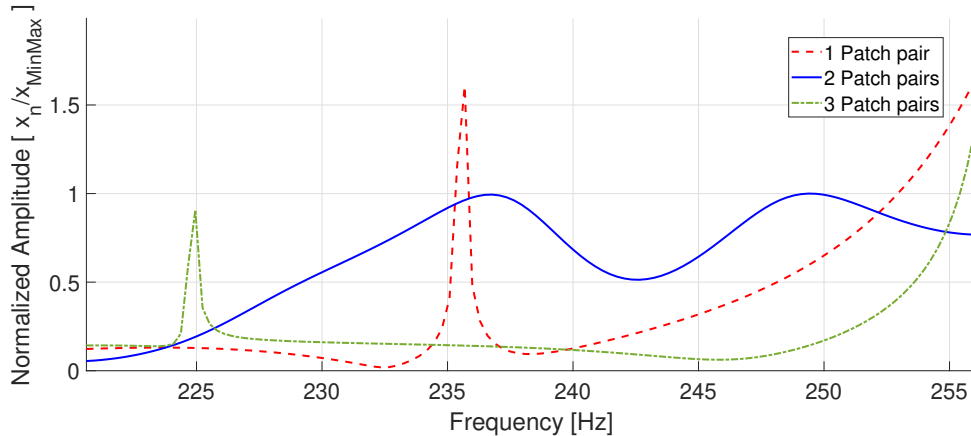


Figure 4.20: Comparison of the best controls implemented for each configuration of patch pair numbers: single-shunt control (unimodal approach) with 1 patch pair, multi-shunt control (multimodal approach) with 2 patch pairs, and single-shunt control (unimodal approach) with 3 patch pairs.

From the curves shown in Figure 4.20, it can be observed that the configuration with 2 patch pairs using the multi-shunt control, optimized via the multimodal approach, exhibited the best overall performance, combining greater stability with the lowest maximum amplitude within the analyzed frequency range. The configuration with 3 patch pairs also demonstrated satisfactory performance; however, as it involves a single-shunt control, its multimodal attenuation capability was limited. Considering these results, Section 4.4 is dedicated to investigating why the single-shunt approach with 3 patch pairs performed better than the multi-shunt approach for this same configuration. This discrepancy may be

related to the effect of the piezoelectric mass distribution on the structure, which alters the harmonic response and can favor certain geometric arrangements. Thus, it is possible that the observed superior performance is more associated with the spatial configuration of the patches than with the effectiveness of the single-shunt control. These aspects are analyzed in greater detail in the following chapter.

4.4 Influence of Mass Distribution

As previously discussed, the results obtained for the shunt control with 3 pairs of piezoelectric patches did not meet expectations, since the single shunt control using the unimodal approach demonstrated better performance — based on the criterion of maximum amplitude reduction — than the multi shunt control with the multimodal approach. This result is counterintuitive, as the use of three patches with independent circuits should, in theory, provide greater attenuation capability, especially when aiming to act on three distinct vibration modes.

The main reason for this unexpected behavior appears to be related to the influence of the mass distribution of the patches on the structure. This distribution alters the system's harmonic response, modifying both the position and magnitude of the vibration peaks. Consequently, certain patch placement positions can, by themselves, lead to a natural reduction in the amplitude of specific modes, reducing the demand for the control circuit to act on these peaks.

To investigate this effect, comparative analyses were carried out between the single shunt control (unimodal approach) and multi shunt control (multimodal approach), focusing specifically on the uncontrolled harmonic curves corresponding to the optimal positions of each approach. The objective of these analyses is to understand the impact of mass distribution on the system's uncontrolled response and, from this, to more fairly evaluate the effectiveness of each control strategy in its respective structural scenario.

4.4.1 Uncontrolled Response

The first analysis aims to verify whether changing the patch positions actually influences the location and magnitude of the system's vibration peaks. Figure 4.21 shows the system's harmonic response without control, considering the patch positions corresponding to the optimal solutions obtained for the unimodal and multimodal approaches, which are described in Tables 4.3 (optimal position based on mode 7) and 4.6 (position defined by the harmonic mean criterion), respectively.

It can be observed in Figure 4.21 that the two previously proposed hypotheses are confirmed: both the positioning of the vibration peaks and their amplitudes are significantly influenced by the placement of the piezoelectric masses. An example of this occurs in mode 6, whose peak appears around 225 Hz. It is noted that, in the configuration optimized using the unimodal approach, the amplitude of this peak is considerably lower than in the multimodal configuration. This implies that, when control is applied, the contribution of this peak to the maximum amplitude becomes practically negligible. In this context, the single shunt control focused on mode 7 proves to be more effective — not necessarily because it

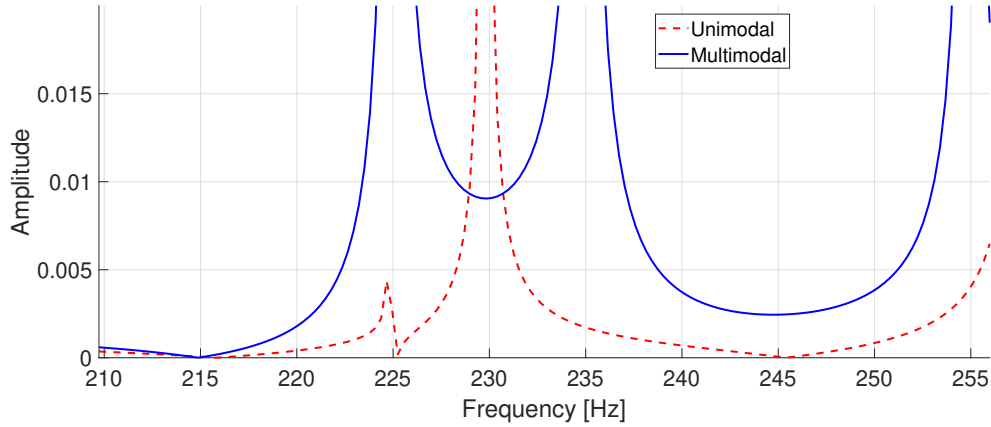


Figure 4.21: System harmonic response without control, considering the optimal patch positioning configurations for the unimodal and multimodal approaches.

attenuates vibration more, but because, due to the natural response of the uncontrolled system, mode 6 already has a lower influence. Therefore, the superior performance of the unimodal approach in this case arises from a passive structural characteristic, and not exclusively from the superiority of the control method itself.

4.4.2 Controlled Response

In order to evaluate the performance of the controls while taking into account the uncontrolled responses specific to each configuration, two complementary analyses were conducted. The first analyzes the absolute reduction of vibration amplitude at each point in the frequency range. The second estimates the vibration energy effectively dissipated by each control approach relative to the potential actuation offered by the adopted configuration.

Amplitude Reduction

In this analysis, the point-by-point difference between the curves with and without control was calculated along the analyzed frequency range, allowing the identification, in absolute terms, of which control resulted in a greater reduction of vibration amplitude. Figure 4.22 presents this comparison for the controls applied with 3 pairs of patches, using RLs calibrated by the unimodal approach and RLs optimized by the multimodal approach.

From the figure, it can be observed that the control based on the multimodal approach provides more significant and distributed reductions for modes 6, 7, and 8. This is due to the nature of the multi-shunt control, which allows tuning different circuits to act simultaneously on multiple modes. On the other hand, the unimodal approach, with single shunt control focused on mode 7, shows a significant reduction only at this specific point, with limited impact on the other modes.

It is important, however, to highlight that the absolute reduction values are directly conditioned by the magnitude of the peaks in the uncontrolled response — which, as discussed previously, vary according

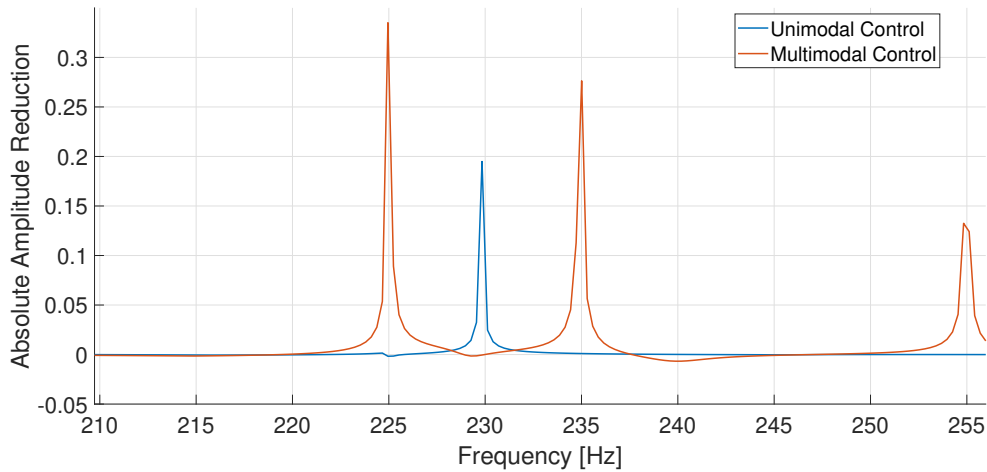


Figure 4.22: Absolute amplitude reductions when applying controls optimized using the unimodal and multimodal approaches, with 3 pairs of piezoelectric patches.

to the mass distribution in the structure. For a fairer comparison, independent of these initial conditions, the next analysis considers the effects of the controls based on a normalized energy criterion.

Dissipated Energy

The second analysis aims to quantify the vibrational energy effectively attenuated by each control strategy. The energy available for actuation is approximately estimated by the area under the “no control” system response curve. Considering that the dissipated energy corresponds to the area between this curve and the “with control” response — the larger this area, the more effective the control is at attenuating structural vibrations. Figure 4.23 graphically illustrates these areas for the unimodal and multimodal approaches, allowing a visual comparison of the energy reduced by each approach.

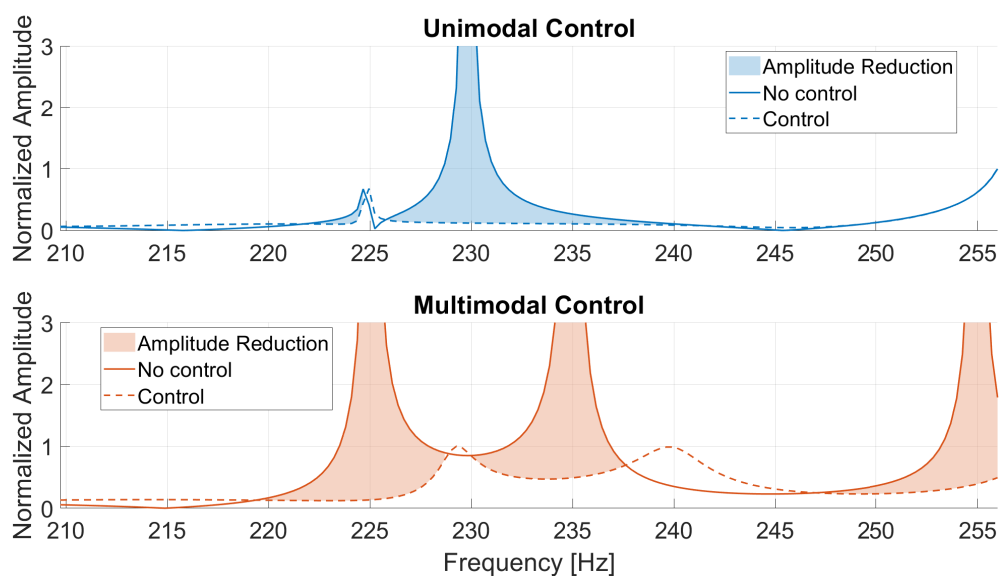


Figure 4.23: Estimated energy dissipated using controls optimized by the unimodal and multimodal approaches, with 3 pairs of piezoelectric patches.

For a more objective comparison, the dissipated energy was normalized relative to the total energy of the uncontrolled response. Figure 4.24 presents the cumulative percentage of energy dissipated across the frequency range for each control approach. This percentage was obtained by calculating the accumulated area between the "no control" and "with control" curves up to each frequency, divided by the total area under the "no control" curve, representing the total vibrational energy available for dissipation. Thus, an ideal control capable of completely attenuating all system energy would make the curve reach 100% at the end of the frequency range, indicating that all available energy has been dissipated.

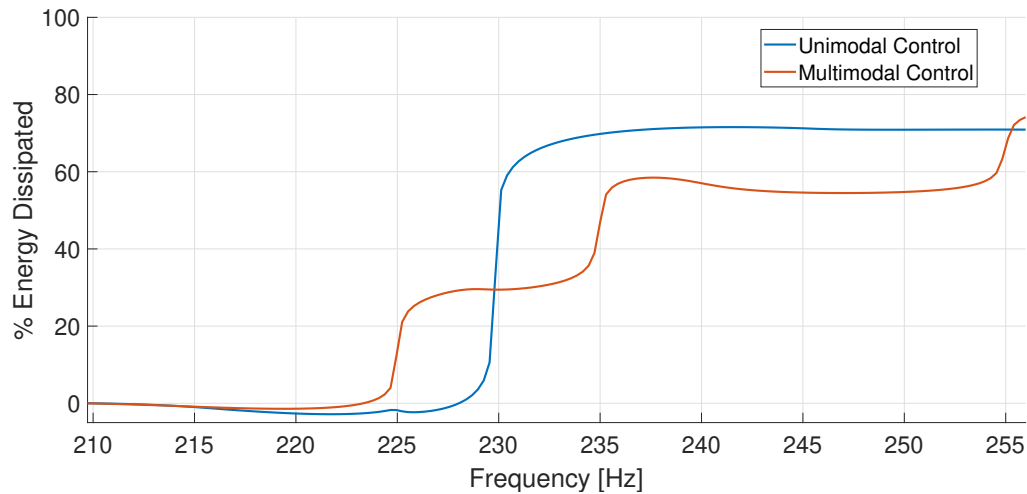


Figure 4.24: Energy efficiency of controls optimized by the unimodal and multimodal approaches, with 3 pairs of piezoelectric patches.

The results show that the multimodal approach is significantly more effective, dissipating a larger fraction of the total vibrational energy compared to the unimodal approach. Moreover, the multimodal curve in Figure 4.24 exhibits three distinct plateaus, indicating significant actuation in three separate modes — consistent with the objective of the multi-shunt control. In contrast, the unimodal approach curve shows only a single step, confirming that the energy reduction was concentrated exclusively in one mode.

These findings reinforce that, although the unimodal control showed better performance in terms of maximum vibration amplitude reduction — conditioned by the specific structural configuration — the multimodal control is clearly superior from a global energy perspective, providing broader and more consistent attenuation of the structure's vibrational modes.

Chapter 5

Conclusions

As discussed throughout this thesis, the use of composite structures has become increasingly common in industry, aiming to reduce structural mass while simultaneously providing desirable mechanical properties. However, one of the main disadvantages of these structures is their susceptibility to structural instabilities. Therefore, it is necessary to employ vibration control techniques, among which passive shunt control stands out as the central focus of this work.

This technique relies on the application of piezoelectric patches on the surface of the structure, which, when connected to electrical elements such as resistors and inductors, enable the dissipation of vibrational energy. This results in the attenuation of vibration peaks and, consequently, the mitigation of structural instabilities. However, its implementation involves several challenges, both geometrically — such as defining the number and distribution of patches — and electrically, including the selection of circuit type and the proper determination of resistance and inductance values.

This thesis focused on a comparative study between single-shunt and multi-shunt controls. The former was developed based on a unimodal approach, in which the patches were positioned to optimize the response of a single vibration mode, with RL parameters determined from calibration equations. The latter employed a multimodal approach, simultaneously considering all relevant modes within the analyzed frequency range. In this case, both the patch positions and the electrical parameters of the circuits were optimized through computational techniques, aiming to minimize the amplitude peaks of the harmonic response.

5.1 Achievements

Based on the results presented in the previous chapter, it can be stated that all objectives defined in Section 1.3 were successfully achieved.

In the initial stage of the research, the control technique was applied using a unimodal approach, analyzing both low- and high-frequency vibration modes. It was found that for low-frequency modes—where modal peaks are well spaced—the unimodal approach proved effective in attenuating vibrations. Conversely, for high-frequency modes, where peaks are closer together, the effectiveness of the single-shunt

control was significantly limited due to its inability to selectively act on multiple modes simultaneously.

Additionally, the results indicated that using more than three pairs of piezoelectric patches not only failed to provide further gains but also compromised system performance. Specifically, the application of four or five pairs showed a tendency for the Electromechanical Coupling Coefficient (EMCC) to decrease, implying reduced efficiency in energy transfer between the mechanical system and the electrical circuit. Furthermore, subdividing a single pair of patches into multiple smaller pairs was found to be a non-advantageous strategy for single-shunt control, as this practice can reduce the collected modal charge—due to the decreased effective area of the patches—and, depending on the excited mode, may even cause signal cancellation, thereby impairing sensor performance and, consequently, the control.

Thus, the main findings regarding the application of calibrated single-shunt control with a unimodal approach can be summarized as follows:

1. The technique is recommended for situations in which only one dominant vibration mode is excited within the frequency range of interest.
2. The use of a single pair of piezoelectric patches is generally sufficient for vibration control, as subdividing into multiple pairs does not improve effectiveness and may even reduce it.
3. The EMCC tends to increase with up to three pairs of patches but decreases when four or more pairs are used, compromising control performance.

Given the limitations observed in the unimodal approach, the research was extended to study the multi-shunt technique using a multimodal approach for high-frequency modes. The results demonstrate that this approach is significantly more effective in attenuating vibrations, primarily due to the possibility of using independent circuits for each pair of patches. This configuration allows each circuit to be tuned to act on specific regions of the frequency spectrum, providing broader, more selective, and more efficient control of the structural response.

However, an important aspect identified during the study relates to the influence of mass distribution caused by patch subdivision. This redistribution can significantly alter both the positions and magnitudes of vibration peaks in the structure. In certain cases, a control may appear more effective simply because the new structural configuration, with redistributed patches, naturally reduces the maximum amplitudes—not necessarily due to the control action itself. To address this issue more rigorously, an analysis based on dissipated vibrational energy was conducted to more accurately assess the effectiveness of the controls applied with three pairs of patches.

The results of this analysis indicated that, although the multi-shunt control may show slightly higher maximum amplitudes compared to single-shunt control, it is more effective from an energy perspective. In other words, the multimodal control demonstrated a greater capacity for vibrational energy dissipation, representing a more robust and globally efficient performance.

Based on the results obtained for the multi-shunt technique with a multimodal approach, the following conclusions can be highlighted:

1. The technique is recommended for applications where multiple vibration modes are present within the frequency range of interest.
2. The adoption of multi-shunt circuits allows for better results in both energy dissipation and maximum amplitude reduction.
3. Subdividing into multiple pairs of patches, when properly positioned, positively contributes to control performance.
4. For systems with three or more pairs of patches, the mass distribution in the structure becomes a critical factor, especially when performance criteria are based on minimizing the maximum amplitude.

5.2 Future Work

Considering the topics discussed previously, the multi-shunt technique proved to be the most promising and efficient approach for vibration control in structures with multiple excited modes. Within the scope of this thesis, the methodology adopted for its implementation was based on two independent steps: the first consisted of a modal analysis, in which the optimal positions of the patches were determined based on the EMCC; the second involved a harmonic analysis, in which, once the positions were fixed, the ideal values of the RL components were calculated.

However, the results obtained with three pairs of patches highlighted an important limitation of this sequential approach: the mass distribution introduced by the piezoelectric patches significantly influences the system's dynamic response, altering both the frequencies and amplitudes of the modal peaks. This compromises the independence between the modal and harmonic analyses, indicating that a truly effective optimized solution requires an integrated approach.

A promising alternative would be the formulation of a unified optimization problem, in which the variables associated with the patch positions and the electrical circuit parameters are considered simultaneously. However, this approach entails high computational costs—both in terms of processing time and memory usage—which is why it was not pursued in this research. This complexity is primarily due to the exponential increase in the search space (with the doubling of the number of variables) and the need to perform complete harmonic simulations for each possible parameter combination.

Therefore, as future work, it is proposed to explore integrated optimization methodologies, ranging from structural simplifications to approaches capable of simulating this integration. Such strategies have the potential to significantly enhance the performance of passive multi-shunt control systems, making them more robust and efficient for real-world applications.

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